

Part IA — Groups

Based on lectures by J. Goedecke

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These notes are not endorsed by the lecturers, and I have modified them (often significantly) after lectures. They are nowhere near accurate representations of what was actually lectured, and in particular, all errors are almost surely mine.

Examples of groups

Axioms for groups. Examples from geometry: symmetry groups of regular polygons, cube, tetrahedron. Permutations on a set; the symmetric group. Subgroups and homomorphisms. Symmetry groups as subgroups of general permutation groups. The Möbius group; cross-ratios, preservation of circles, the point at infinity. Conjugation. Fixed points of Möbius maps and iteration. [4]

Lagrange's theorem

Cosets. Lagrange's theorem. Groups of small order (up to order 8). Quaternions. Fermat-Euler theorem from the group-theoretic point of view. [5]

Group actions

Group actions; orbits and stabilizers. Orbit-stabilizer theorem. Cayley's theorem (every group is isomorphic to a subgroup of a permutation group). Conjugacy classes. Cauchy's theorem. [4]

Quotient groups

Normal subgroups, quotient groups and the isomorphism theorem. [4]

Matrix groups

The general and special linear groups; relation with the Möbius group. The orthogonal and special orthogonal groups. Proof (in $\mathbb{R}^3$) that every element of the orthogonal group is the product of reflections and every rotation in $\mathbb{R}^3$ has an axis. Basis change as an example of conjugation. [3]

Permutations

Permutations, cycles and transpositions. The sign of a permutation. Conjugacy in S_n and in A_n . Simple groups; simplicity of A_5 . [4]

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0 Introduction

Group theory is an example of *algebra*. In pure mathematics, algebra (usually) does not refer to the boring mindless manipulation of symbols. Instead, in algebra, we have some set of objects with some operations on them. For example, we can take the integers with addition as the operation. However, in algebra, we allow *any* set and *any* operations, not just numbers.

Of course, such a definition is too broad to be helpful. We categorize algebraic structures into different types. In this course, we will study a particular kind of structures, *groups*. In the IB Groups, Rings and Modules course, we will study rings and modules as well.

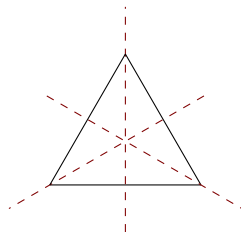
These different kinds of structures are defined by certain *axioms*. The *group axioms* will say that the operation must follow certain rules, and any set and operation that satisfies these rules will be considered to form a group. We will then have a different set of axioms for rings, modules etc.

As mentioned above, the most familiar kinds of algebraic structures are number systems such as integers and rational numbers. The focus of group theory, however, is not on things that resemble “numbers”. Instead, it is the study of *symmetries*.

First of all, what is a symmetry? We are all familiar with, say, the symmetries of an (equilateral) triangle (we will always assume the triangle is equilateral). We rotate a triangle by 120° , and we get the original triangle. We say that rotating by 120° is a symmetry of a triangle. In general, a symmetry is something we do to an object that leaves the object intact.

Of course, we don’t require that the symmetry leaves *everything* intact. Otherwise, we would only be allowed to do nothing. Instead, we require certain important things to be intact. For example, when considering the symmetries of a triangle, we only care about how the resultant object looks, but don’t care about where the individual vertices went.

In the case of the triangle, we have six symmetries: three rotations (rotation by 0° , 120° and 240°), and three reflections along the axes below:



These six together form the underlying set of the *group of symmetries*. A more sophisticated example is the symmetries of $\mathbb{R}^3$. We define these as operations on $\mathbb{R}^3$ that leave distances between points unchanged. These include translations, rotations, reflections, and combinations of these.

So what is the operation? This operation combines two symmetries to give a new symmetry. The natural thing to do is to do the symmetry one after another. For example, if we combine the two 120° rotations, we get a 240° rotation.

Now we are studying algebra, not geometry. So to define the group, we *abstract away* the triangle. Instead, we define the group to be six objects, say $\{e, r, r^2, s, rs, r^2s\}$, with rules defining how we combine two elements to get a third. Officially, we do not mention the triangle at all when defining the group.

We can now come up with the group axioms. What rules should the set of symmetries obey? First of all, we must have a “do nothing” symmetry. We call this the *identity* element. When we compose the identity with another symmetry, the other symmetry is unchanged.

Secondly, given a symmetry, we can do the reverse symmetry. So for any element, there is an inverse element that, when combined with the original, gives the identity.

Finally, given three symmetries, we can combine them, one after another. If we denote the operation of the group as $*$, then if we have three symmetries, x, y, z , we should be able to form $x * y * z$. If we want to define it in terms of the binary operation $*$, we can define it as $(x * y) * z$, where we first combine the first two symmetries, then combine the result with the third. Alternatively, we can also define it as $x * (y * z)$. Intuitively, these two should give the same result, since both are applying x after y after z . Hence we have the third rule $x * (y * z) = (x * y) * z$.

Now a group is any set with an operation that satisfies the three rules above. In group theory, the objective is to study the properties of groups just assuming these three axioms. It turns out that there is a *lot* we can talk about.

1 Groups and homomorphisms

1.1 Groups

Definition (Binary operation). A *(binary) operation* on a set A is a map $*$: $A \times A \rightarrow A$.

Definition (Group). A *group* is a set G together with a binary operation $*$ on G satisfying the following axioms:

1. There exists an element $e \in G$ such that for all $a \in G$,

$$a * e = e * a = a. \quad (\text{identity})$$

2. For all $a \in G$, there exists an element $a^{-1} \in G$ such that

$$a * a^{-1} = a^{-1} * a = e. \quad (\text{inverse})$$

3. For all $a, b, c \in G$,

$$(a * b) * c = a * (b * c). \quad (\text{associativity})$$

Definition (Order of group). The *order* of a group G , denoted by $|G|$, is the number of elements in G . A group is a *finite group* if its order is finite.

Remark. Strictly speaking, the inverse axiom as stated is not yet well-defined, since we have not specified what e is. The identity axiom only guarantees the existence of *some* e satisfying the identity property, but there could be many such elements. Thus, the inverse axiom should be interpreted as: for each $a \in G$, there exists $a^{-1} \in G$ such that $a * a^{-1}$ and $a^{-1} * a$ both satisfy the identity property. We will soon show that the identity is unique, after which this ambiguity is resolved.

Remark. Some authors include a zeroth axiom called *closure*:

0. For all $a, b \in G$, we have $a * b \in G$. (closure)

Since $*$ is defined to be a binary operation on G , this is automatically satisfied. However, in practice, one often needs to verify it explicitly. For example, if we let G be the set of all matrices of the form

$$\begin{pmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix}$$

under matrix multiplication, we must check that the product of two such

matrices is again of this form. Formally, we are verifying that the binary operation is well-defined on G .

Remark. In general, it is *not* the case that $a * b = b * a$ for all elements a, b of a group G . For example, in the group of symmetries of a triangle, rotating and then reflecting is different from reflecting and then rotating.

Definition (Abelian group). A group $(G, *)$ is *abelian* if it additionally satisfies

$$4. (\forall a, b \in G) a * b = b * a. \quad (\text{commutativity})$$

Notation. When the operation $*$ is clear from context, we write ab in place of $a * b$. We also write $a^2 = aa$, $a^n = \underbrace{aaa \cdots a}_{n \text{ copies}}$, $a^0 = e$, and $a^{-n} = (a^{-1})^n$ for $n \geq 1$.

Example. The following are abelian groups:

- (i) $(\mathbb{Z}, +)$
- (ii) $(\mathbb{Q}, +)$
- (iii) $(\mathbb{Z}_n, +_n)$, the integers modulo n under addition modulo n
- (iv) $(\mathbb{Q}^*, \times)$, the nonzero rationals under multiplication
- (v) $(\{-1, 1\}, \times)$

The following are non-abelian groups:

- (vi) The symmetry group of a regular n -gon under composition, denoted D_{2n}
- (vii) The group of 2×2 invertible real matrices under matrix multiplication, denoted $\text{GL}_2(\mathbb{R})$
- (viii) Symmetry groups of 3D objects

Proposition (Uniqueness of identity and inverse). Let $(G, *)$ be a group. Then

- (i) The identity element is unique.
- (ii) For each $a \in G$, the inverse of a is unique.

Proof.

- (i) Suppose e and e' are both identity elements. Then $ee' = e'$, using the fact

that e is an identity, and $ee' = e$, using the fact that e' is an identity. Thus $e = e'$.

- (ii) Suppose a^{-1} and b both satisfy the inverse axiom for some $a \in G$. Then $b = be = b(aa^{-1}) = (ba)a^{-1} = ea^{-1} = a^{-1}$. Thus $b = a^{-1}$. $\square$

Remark. Since the identity is unique, it is now meaningful to speak of *the* identity element e . Similarly, since inverses are unique, we may unambiguously write a^{-1} for *the* inverse of a .

Proposition (Inverse laws). Let $(G, *)$ be a group and let $a, b \in G$. Then

- (i) $(a^{-1})^{-1} = a$.
(ii) $(ab)^{-1} = b^{-1}a^{-1}$.

Proof.

- (i) Given a^{-1} , both a and $(a^{-1})^{-1}$ satisfy

$$xa^{-1} = a^{-1}x = e.$$

By uniqueness of inverses, $(a^{-1})^{-1} = a$.

- (ii) We have

$$\begin{aligned} (ab)(b^{-1}a^{-1}) &= a(bb^{-1})a^{-1} \\ &= aea^{-1} \\ &= aa^{-1} \\ &= e \end{aligned}$$

Similarly, $(b^{-1}a^{-1})ab = e$. So $b^{-1}a^{-1}$ is an inverse of ab . By the uniqueness of inverses, $(ab)^{-1} = b^{-1}a^{-1}$. $\square$

Definition (Subgroup). Let $(G, *)$ be a group. A subset $H \subseteq G$ is a *subgroup* of G , written $H \leq G$, if H with the restricted operation $*$ from G is itself a group.

Example.

- $(\mathbb{Z}, +) \leq (\mathbb{Q}, +) \leq (\mathbb{R}, +) \leq (\mathbb{C}, +)$.
- $(\{e\}, *) \leq (G, *)$ for any group G (the *trivial subgroup*).
- $G \leq G$.
- $(\{\pm 1\}, \times) \leq (\mathbb{Q}^*, \times)$.

Remark. To show that H is a subgroup of G directly from the definition, one must verify all the group axioms for H . The following two lemmas provide more convenient criteria.

Lemma (Subgroup criteria I). Let $(G, *)$ be a group and let $H \subseteq G$. Then $H \leq G$ if and only if

- (i) $e \in H$;
- (ii) $(\forall a, b \in H) ab \in H$;
- (iii) $(\forall a \in H) a^{-1} \in H$.

Proof. If $H \leq G$, then (i), (ii), and (iii) hold by the group axioms.

Conversely, suppose (i), (ii), and (iii) hold. The group axioms for H are then verified as follows:

- 0. Closure: by (ii).
- 1. Identity: by (i). Note that H and G must have the same identity. Indeed, suppose e_H and e_G are the identities of H and G respectively. Then $e_H e_H = e_H$. Since e_H has an inverse e_H^{-1} in G , multiplying on the right gives $e_H = e_G$.
- 2. Inverse: by (iii).
- 3. Associativity: inherited from G . □

Lemma (Subgroup criteria II). Let $(G, *)$ be a group and let $H \subseteq G$. Then $H \leq G$ if and only if

- (I) H is non-empty;
- (II) $(\forall a, b \in H) ab^{-1} \in H$.

Proof. If $H \leq G$, then (I) and (II) follow immediately from (i), (ii), and (iii) of Subgroup criteria I.

Conversely, suppose (I) and (II) hold. We verify conditions (i), (ii), and (iii):

- (i) Since H is non-empty, it contains some element a . Then $aa^{-1} = e \in H$ by (II).
- (iii) For any $a \in H$, we have $ea^{-1} = a^{-1} \in H$ by (II).
- (ii) For any $a, b \in H$, since $b^{-1} \in H$ by (iii), we have $a(b^{-1})^{-1} = ab \in H$ by (II). □

Proposition (Subgroups of $\mathbb{Z}$). The subgroups of $(\mathbb{Z}, +)$ are exactly the sets $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ for $n \in \mathbb{N}$.

Proof. For any $n \in \mathbb{N}$, it is straightforward to verify that $n\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$.

Conversely, let $H \leq \mathbb{Z}$. We have $0 \in H$. If $H = \{0\}$, then $H = 0\mathbb{Z}$ and we are done.

Otherwise, H contains a nonzero element a , and since $-a \in H$ as well, H contains a positive integer. Let n be the smallest positive integer in H . We claim that $H = n\mathbb{Z}$. Since $n \in H$ and H is closed under addition and inverses, all multiples of n belong to H , so $n\mathbb{Z} \subseteq H$. Now suppose for contradiction that there exists $a \in H$ with $n \nmid a$. By the division algorithm, write $a = pn + q$ where $0 < q < n$. Since $a - pn \in H$, we have $q \in H$. But $0 < q < n$, contradicting the minimality of n . Hence every element of H is divisible by n , so $H \subseteq n\mathbb{Z}$. Therefore $H = n\mathbb{Z}$. $\square$

1.2 Homomorphisms

Remark. To study groups, it is essential to study structure-preserving functions between them. We first recall the basic definitions concerning functions.

Definition (Function). Given two sets X and Y , a *function* $f: X \rightarrow Y$ sends each $x \in X$ to a particular element $f(x) \in Y$. The set X is called the *domain* and Y is called the *codomain*.

Example.

- Identity function: for any set X , $1_X: X \rightarrow X$ with $1_X(x) = x$ is a function. This is also written as id_X .
- Inclusion map: $\iota: \mathbb{Z} \rightarrow \mathbb{Q}$ defined by $\iota(n) = n$. Note that this differs from the identity function, as the domain and codomain are different.
- $f_1: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f_1(x) = x + 1$.
- $f_2: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f_2(x) = 2x$.
- $f_3: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f_3(x) = x^2$.
- Consider functions $\{0, 1, 2, 3, 4\} \rightarrow \{0, 1, 2, 3, 4\}$:
 - $g_1(x) = x + 1$ if $x < 4$; $g_1(4) = 4$.
 - $g_2(x) = x + 1$ if $x < 4$; $g_2(4) = 0$.

Definition (Composition of functions). Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be functions. The *composition* of g with f is the function $g \circ f: X \rightarrow Z$ defined by $(g \circ f)(x) = g(f(x))$.

Example. Using f_1 and f_2 defined above, $(f_2 \circ f_1)(x) = 2x + 2$ and $(f_1 \circ f_2)(x) = 2x + 1$. Note that function composition is not commutative in general.

Definition (Injective function). A function $f: X \rightarrow Y$ is *injective* if distinct elements have distinct images, i.e.

$$(\forall x, y \in X) f(x) = f(y) \Rightarrow x = y.$$

Definition (Surjective function). A function $f: X \rightarrow Y$ is *surjective* if every element of Y is in the image of f , i.e.

$$(\forall y \in Y)(\exists x \in X) f(x) = y.$$

Definition (Bijective function). A function $f: X \rightarrow Y$ is *bijective* if it is both injective and surjective. A function has an inverse if and only if it is bijective.

Example. ι and f_2 are injective but not surjective. f_3 and g_1 are neither. 1_X , f_1 and g_2 are bijective.

Lemma (Composition of bijections). The composition of two bijective functions is bijective.

Remark. When studying groups, we are not interested in arbitrary functions between the underlying sets. Instead, we consider functions that preserve the group structure, called *homomorphisms*.

Definition (Group homomorphism). Let $(G, *)$ and $(H, \times)$ be groups. A function $f: G \rightarrow H$ is a *group homomorphism* if

$$(\forall g_1, g_2 \in G) f(g_1) \times f(g_2) = f(g_1 * g_2).$$

Definition (Group isomorphism). A *group isomorphism* is a bijective group homomorphism. Two groups G and H are *isomorphic*, written $G \cong H$, if there exists an isomorphism between them.

Remark. Isomorphic groups are considered to be “the same” from a group-theoretic perspective. For example, when we say that there is

only one group of order 2, we mean that any two groups of order 2 are isomorphic.

Example.

- For any groups G and H , the function $f: G \rightarrow H$ defined by $f(g) = e_H$ for all $g \in G$ is a homomorphism (the *trivial homomorphism*).
- $1_G: G \rightarrow G$ and $f_2: \mathbb{Z} \rightarrow 2\mathbb{Z}$ are isomorphisms. $\iota: \mathbb{Z} \rightarrow \mathbb{Q}$ and $f_2: \mathbb{Z} \rightarrow \mathbb{Z}$ are homomorphisms.
- $\exp: (\mathbb{R}, +) \rightarrow (\mathbb{R}^+, \times)$ defined by $\exp(x) = e^x$ is an isomorphism.
- Let $H = (\{e^{ik\pi/2} : k = 0, 1, 2, 3\}, \times)$. Then $f: \mathbb{Z}_4 \rightarrow H$ defined by $f(a) = e^{i\pi a/2}$ is an isomorphism.
- $\det: \text{GL}_2(\mathbb{R}) \rightarrow \mathbb{R}^*$ is a homomorphism.

Proposition (Elementary homomorphism properties). Let $f: G \rightarrow H$ be a group homomorphism. Then

- (i) $f(e_G) = e_H$.
- (ii) For all $a \in G$, $f(a^{-1}) = f(a)^{-1}$.
- (iii) The composite of two group homomorphisms is a group homomorphism.
- (iv) The inverse of an isomorphism is an isomorphism.

Proof.

(i)

$$\begin{aligned} f(e_G) &= f(e_G^2) = f(e_G)^2 \\ f(e_G)^{-1}f(e_G) &= f(e_G)^{-1}f(e_G)^2 \\ f(e_G) &= e_H \end{aligned}$$

(ii)

$$\begin{aligned} e_H &= f(e_G) \\ &= f(aa^{-1}) \\ &= f(a)f(a^{-1}) \end{aligned}$$

Since inverses are unique, $f(a^{-1}) = f(a)^{-1}$.

- (iii) Let $f: G_1 \rightarrow G_2$ and $g: G_2 \rightarrow G_3$ be homomorphisms. Then $g(f(ab)) = g(f(a)f(b)) = g(f(a))g(f(b))$.

(iv) Let $f: G \rightarrow H$ be an isomorphism and let $a, b \in H$. Then

$$\begin{aligned} f^{-1}(ab) &= f^{-1}\left\{f[f^{-1}(a)]f[f^{-1}(b)]\right\} \\ &= f^{-1}\left\{f[f^{-1}(a)f^{-1}(b)]\right\} \\ &= f^{-1}(a)f^{-1}(b) \end{aligned}$$

So f^{-1} is a homomorphism. Since it is bijective, f^{-1} is an isomorphism. $\square$

Definition (Image of homomorphism). Let $f: G \rightarrow H$ be a group homomorphism. The *image* of f is

$$\text{im } f = f(G) = \{f(g) : g \in G\}.$$

Definition (Kernel of homomorphism). Let $f: G \rightarrow H$ be a group homomorphism. The *kernel* of f is

$$\ker f = f^{-1}(\{e_H\}) = \{g \in G : f(g) = e_H\}.$$

Proposition (Image and kernel subgroup property). Let $f: G \rightarrow H$ be a group homomorphism. Then $\text{im } f \leq H$ and $\ker f \leq G$.

Proof. Since $f(e_G) = e_H$, we have $e_H \in \text{im } f$ and $e_G \in \ker f$, so both sets are non-empty. For the image, suppose $b_1, b_2 \in \text{im } f$, so there exist $a_1, a_2 \in G$ with $f(a_1) = b_1$ and $f(a_2) = b_2$. Then $b_1 b_2^{-1} = f(a_1)f(a_2)^{-1} = f(a_1 a_2^{-1}) \in \text{im } f$.

For the kernel, suppose $b_1, b_2 \in \ker f$. Then $f(b_1 b_2^{-1}) = f(b_1)f(b_2)^{-1} = e_H e_H^{-1} = e_H$, so $b_1 b_2^{-1} \in \ker f$. $\square$

Proposition (Kernel conjugation closure). Let $f: G \rightarrow H$ be a group homomorphism. Then for all $a \in G$ and all $k \in \ker f$, we have $aka^{-1} \in \ker f$.

Proof. $f(aka^{-1}) = f(a)f(k)f(a)^{-1} = f(a)e_H f(a)^{-1} = e_H$. So $aka^{-1} \in \ker f$. $\square$

Remark. The above property—that $aka^{-1} \in \ker f$ for all $a \in G$ and $k \in \ker f$ —is not special to kernels. Subgroups satisfying this property are called *normal subgroups* and play a central role in the theory, as we will see later.

Example. Images and kernels for previously defined homomorphisms:

- (i) For the trivial homomorphism $f: G \rightarrow H$ with $f(g) = e_H$, we have $\text{im } f = \{e_H\}$ and $\ker f = G$.
- (ii) For the identity function, $\text{im } 1_G = G$ and $\ker 1_G = \{e\}$.
- (iii) For the inclusion map $\iota: \mathbb{Z} \rightarrow \mathbb{Q}$, we have $\text{im } \iota = \mathbb{Z}$ and $\ker \iota = \{0\}$.

- (iv) For $f_2: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f_2(x) = 2x$, we have $\text{im } f_2 = 2\mathbb{Z}$ and $\ker f_2 = \{0\}$.
- (v) For $\det: \text{GL}_2(\mathbb{R}) \rightarrow \mathbb{R}^*$, we have $\text{im } \det = \mathbb{R}^*$ and $\ker \det = \{A \in \text{GL}_2(\mathbb{R}) : \det A = 1\} = \text{SL}_2(\mathbb{R})$.

Proposition (Injectivity and surjectivity criterion). Let $f: G \rightarrow H$ be a group homomorphism. Then f is

- (i) surjective if and only if $\text{im } f = H$;
- (ii) injective if and only if $\ker f = \{e_G\}$.

Proof.

- (i) This follows directly from the definitions.
- (ii) Since $f(e_G) = e_H$, we have $e_G \in \ker f$. If f is injective, then e_G is the only element mapping to e_H , so $\ker f = \{e_G\}$. Conversely, if $\ker f = \{e_G\}$, then given $a, b \in G$ with $f(a) = f(b)$, we have $f(ab^{-1}) = f(a)f(b)^{-1} = e_H$. Thus $ab^{-1} \in \ker f = \{e_G\}$, so $ab^{-1} = e_G$ and hence $a = b$. $\square$

Remark. The image and kernel of a homomorphism are not merely convenient terminology. They are related by the *first isomorphism theorem*, which we will prove after developing some further theory.

1.3 Cyclic groups

Remark. The simplest class of groups is that of *cyclic groups*. Informally, a cyclic group is a group of the form $\{e, a, a^2, a^3, \dots, a^{n-1}\}$, where $a^n = e$. For example, the group of rotations of an equilateral triangle is $\{e, r, r^2\}$ where r denotes rotation by 120° and $r^3 = e$.

Definition (Cyclic group). A group G is *cyclic* if

$$(\exists a \in G)(\forall b \in G)(\exists n \in \mathbb{Z}) b = a^n,$$

i.e. every element of G is a power of a . Such an element a is called a *generator* of G .

We write C_n for the cyclic group of order n .

Example.

- (i) $(\mathbb{Z}, +)$ is cyclic with generator 1 (or -1). It is *the* infinite cyclic group.
- (ii) $(\{+1, -1\}, \times)$ is cyclic with generator -1 .

(iii) $(\mathbb{Z}_n, +)$ is cyclic; its generators are precisely the elements coprime to n .

Notation. Given a group G and an element $a \in G$, we write $\langle a \rangle$ for the *cyclic subgroup generated by a* , i.e. the subgroup of all powers of a . It is the smallest subgroup of G containing a .

Definition (Order of element). Let G be a group and let $a \in G$. The *order* of a , denoted $\text{ord}(a)$, is the smallest positive integer n such that $a^n = e$. If no such n exists, we say a has *infinite order*.

Remark. The word “order” now has two meanings: the order of a group ($|G|$) and the order of an element ($\text{ord}(a)$). The following lemma shows that these are closely related.

Lemma (Order equals cyclic subgroup size). Let G be a group and let $a \in G$. Then $\text{ord}(a) = |\langle a \rangle|$.

Proof. If $\text{ord}(a) = \infty$, then $a^n \neq a^m$ for all $n \neq m$ (otherwise $a^{m-n} = e$, contradicting infinite order). Thus $|\langle a \rangle| = \infty = \text{ord}(a)$.

Otherwise, suppose $\text{ord}(a) = k$, so that $a^k = e$. We claim that $\langle a \rangle = \{e, a, a^2, \dots, a^{k-1}\}$. Since $a^k = e$, every power of a reduces to one of $e, a, \dots, a^{k-1}$, so $\langle a \rangle$ contains no additional elements. Moreover, these k elements are distinct: if $a^m = a^n$ for $0 \leq n < m \leq k-1$, then $a^{m-n} = e$ with $0 < m-n < k$, contradicting the minimality of k . Hence $|\langle a \rangle| = k = \text{ord}(a)$. $\square$

Proposition (Cyclic group commutativity). Every cyclic group is abelian.

Proof. Let $G = \langle a \rangle$ be cyclic. For any $b, c \in G$, we have $b = a^m$ and $c = a^n$ for some $m, n \in \mathbb{Z}$. Then $bc = a^m a^n = a^{m+n} = a^n a^m = cb$. $\square$

Definition (Exponent of group). The *exponent* of a group G is the smallest positive integer n such that $a^n = e$ for all $a \in G$.

1.4 Dihedral groups

Definition (Dihedral group D_{2n}). For $n \geq 3$, the *dihedral group* D_{2n} is the group of symmetries of a regular n -gon. It has order $2n$, consisting of n rotations (including the identity) and n reflections, and admits the

presentation

$$\begin{aligned} D_{2n} &= \langle r, s \mid r^n = s^2 = e, srs^{-1} = r^{-1} \rangle \\ &= \{e, r, r^2, \dots, r^{n-1}, s, rs, r^2s, \dots, r^{n-1}s\}. \end{aligned}$$

The subscript in D_{2n} refers to the order of the group, not the number of sides of the polygon.

Remark. Geometrically, r represents rotation by $360^\circ/n$ (which has order n) and s represents a fixed reflection (which has order 2). The relation $srs^{-1} = r^{-1}$ expresses the fact that conjugating a rotation by a reflection reverses the direction of rotation. Every element of D_{2n} can be written uniquely in the form r^m (a rotation) or r^ms (a reflection) for $0 \leq m \leq n-1$.

Notation. The notation $\langle \text{generators} \mid \text{relations} \rangle$ is called a *group presentation*. For example, the cyclic group of order n can be written as

$$C_n = \langle a \mid a^n = e \rangle.$$

1.5 Direct products of groups

Definition (Direct product of groups). Given two groups $(G, \circ)$ and $(H, \bullet)$, the *direct product* $G \times H$ is the set of ordered pairs $\{(g, h) : g \in G, h \in H\}$ equipped with the componentwise operation

$$(a_1, a_2) * (b_1, b_2) = (a_1 \circ b_1, a_2 \bullet b_2).$$

This is a group with identity (e_G, e_H) and inverses $(g, h)^{-1} = (g^{-1}, h^{-1})$.

Remark. The direct product arises naturally when considering symmetries of independent objects. For example, the symmetry group of a system of two independent equilateral triangles is $D_6 \times D_6$.

Example.

$$\begin{aligned} C_2 \times C_2 &= \{(0, 0), (0, 1), (1, 0), (1, 1)\} \\ &= \{e, x, y, xy\} \text{ with every non-identity element of order 2} \\ &= \langle x, y \mid x^2 = y^2 = e, xy = yx \rangle. \end{aligned}$$

Proposition (Cyclic direct product criterion). $C_n \times C_m \cong C_{nm}$ if and only if $\text{hcf}(m, n) = 1$.

Proof. Suppose that $\text{hcf}(m, n) = 1$. Let $C_n = \langle a \rangle$ and $C_m = \langle b \rangle$, and let $k = \text{ord}((a, b))$ in $C_n \times C_m$. Then $(a, b)^k = (a^k, b^k) = (e, e)$, which requires $n \mid k$ and $m \mid k$, i.e. k is a common multiple of n and m . Since k is the smallest such positive integer, $k = \text{lcm}(n, m) = \frac{nm}{\text{hcf}(n, m)} = nm$.

Now $\langle (a, b) \rangle \leq C_n \times C_m$, and since (a, b) has order $nm = |C_n \times C_m|$, we have $\langle (a, b) \rangle = C_n \times C_m$. Thus $C_n \times C_m$ is cyclic of order nm , so $C_n \times C_m \cong C_{nm}$.

Conversely, suppose $\text{hcf}(m, n) \neq 1$. Let $k = \text{lcm}(m, n) < mn$. Then for any $(a, b) \in C_n \times C_m$, we have $(a, b)^k = (a^k, b^k) = (e, e)$, so the order of every element divides $k < mn$. In particular, no element has order mn , so $C_n \times C_m$ is not cyclic of order mn . $\square$

Remark. Given a group G , it is sometimes useful to recognise it as a direct product of two of its subgroups. The following proposition gives a criterion for doing so.

Proposition (Direct product theorem). Let G be a group and let $H_1, H_2 \leq G$. Suppose:

- (i) $H_1 \cap H_2 = \{e\}$;
- (ii) $(\forall a_1 \in H_1)(\forall a_2 \in H_2) a_1 a_2 = a_2 a_1$;
- (iii) $(\forall a \in G)(\exists a_1 \in H_1)(\exists a_2 \in H_2) a = a_1 a_2$ (i.e. $G = H_1 H_2$).

Then $G \cong H_1 \times H_2$.

Proof. Define $f: H_1 \times H_2 \rightarrow G$ by $f(a_1, a_2) = a_1 a_2$. This is a homomorphism, since by (ii),

$$\begin{aligned} f((a_1, a_2) * (b_1, b_2)) &= f(a_1 b_1, a_2 b_2) \\ &= a_1 b_1 a_2 b_2 \\ &= a_1 a_2 b_1 b_2 \\ &= f(a_1, a_2) f(b_1, b_2). \end{aligned}$$

Surjectivity follows from (iii). For injectivity, suppose $f(a_1, a_2) = e$, i.e. $a_1 a_2 = e$. Then $a_1 = a_2^{-1}$. Since $a_1 \in H_1$ and $a_2^{-1} \in H_2$, we have $a_1 = a_2^{-1} \in H_1 \cap H_2 = \{e\}$ by (i). Thus $a_1 = a_2 = e$, so $\ker f = \{e\}$ and f is injective. $\square$

2 Symmetric group I

Remark. Recall that a symmetry of an object is a bijection that preserves some structure. For example, a symmetry of $\mathbb{R}^2$ is a bijection $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that preserves distances. We now consider the case where no structure needs to be preserved: the group of *all* bijections from a set to itself. This is the *symmetric group*, and it is in some sense the most general symmetry group. We will later show (in Chapter 5) that every group is isomorphic to a subgroup of some symmetric group.

2.1 Symmetric groups

Definition (Permutation). Let X be a set. A *permutation* of X is a bijection $\sigma: X \rightarrow X$. The set of all permutations of X is denoted $\text{Sym } X$.

Remark. We compose permutations as functions: if σ and τ are permutations, then $\sigma \circ \tau$ denotes first applying τ , then applying σ .

Theorem (Symmetric group). $(\text{Sym } X, \circ)$ is a group.

Proof. The group axioms are verified as follows:

0. If $\sigma, \tau \in \text{Sym } X$, then $\sigma \circ \tau: X \rightarrow X$ is a bijection, so $\sigma \circ \tau \in \text{Sym } X$.
1. The identity function $1_X: X \rightarrow X$ is a permutation and satisfies $\sigma \circ 1_X = 1_X \circ \sigma = \sigma$ for all $\sigma \in \text{Sym } X$.
2. Every bijection has a bijective inverse, so if $\sigma \in \text{Sym } X$, then $\sigma^{-1} \in \text{Sym } X$.
3. Composition of functions is associative. □

Definition (Symmetric group S_n). If X is a finite set with $|X| = n$ (we usually take $X = \{1, 2, \dots, n\}$), we write $S_n = \text{Sym } X$. This is the *symmetric group* of degree n .

Remark. The *degree* of S_n is n , while the *order* of S_n is $|S_n| = n!$. For example, S_3 has degree 3 but order 6.

Notation (Two row notation). A permutation $\sigma \in S_n$ can be written by listing $1, 2, \dots, n$ on the top row and their images below:

$$\begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \cdots & \sigma(n) \end{pmatrix}.$$

For example,

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \in S_3 \quad \text{and} \quad \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 3 & 4 & 5 \end{pmatrix} \in S_5.$$

Example. For small values of n :

$$(i) \quad S_1 = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} = \{e\} \cong C_1.$$

$$(ii) \quad S_2 = \left\{ \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \right\} \cong C_2.$$

$$(iii) \quad S_3 = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \right\} \cong D_6.$$

Remark. S_3 is not abelian, and hence S_n is not abelian for any $n \geq 3$, since S_3 can be viewed as a subgroup of S_n (by fixing $4, 5, \dots, n$). More generally, D_{2n} can be viewed as a subgroup of S_n , since each symmetry of a regular n -gon permutes its vertices.

Notation (Cycle notation). A permutation that sends $a_1 \mapsto a_2 \mapsto \dots \mapsto a_k \mapsto a_1$ and fixes all other elements is written as the cycle $(a_1 \ a_2 \ \dots \ a_k)$. By convention, we write the smallest element first and omit fixed points. More general permutations can be written as products of cycles; for example, $(1 \ 2)(3 \ 4) \in S_4$ denotes the permutation that swaps 1 with 2 and 3 with 4.

Proposition (Cycle inverse). The inverse of a cycle is obtained by writing it in reverse order, e.g. $(1 \ 2 \ 3)^{-1} = (3 \ 2 \ 1) = (1 \ 3 \ 2)$.

Example.

$$(i) \quad (1 \ 2 \ 3)(1 \ 2) = (1 \ 3), \text{ since (composing right to left) } 1 \mapsto 2 \mapsto 3, \\ 3 \mapsto 3 \mapsto 1, \text{ and } 2 \mapsto 1 \mapsto 2.$$

$$(ii) \quad (1 \ 2 \ 3 \ 4)(1 \ 4) = (2 \ 3 \ 4).$$

Definition (k -cycles and transpositions). A cycle $(a_1 \ a_2 \ \dots \ a_k)$ is called a k -cycle. A 2-cycle is called a *transposition*. Two cycles are *disjoint* if no element appears in both.

Example. $(1\ 2)$ and $(3\ 4)$ are disjoint, but $(1\ 2\ 3)$ and $(1\ 2)$ are not.

Lemma (Disjoint cycle commutativity). Disjoint cycles commute.

Proof. Let $\sigma, \tau \in S_n$ be disjoint cycles. We show that $\sigma(\tau(a)) = \tau(\sigma(a))$ for all $a \in \{1, \dots, n\}$. If a appears in neither σ nor τ , then both sides equal a . Otherwise, assume without loss of generality that a appears in τ but not in σ . Then $\tau(a)$ also appears in τ and hence not in σ , so $\sigma(\tau(a)) = \tau(a)$. On the other hand, $\sigma(a) = a$, so $\tau(\sigma(a)) = \tau(a)$. Thus $\sigma \circ \tau = \tau \circ \sigma$. $\square$

Remark. Non-disjoint cycles need not commute. For example, $(1\ 3)(2\ 3) = (1\ 3\ 2)$ while $(2\ 3)(1\ 3) = (1\ 2\ 3)$.

Theorem (Disjoint cycle decomposition). Every permutation in S_n can be written as a product of disjoint cycles. This decomposition is unique up to the ordering of cycles and cyclic rotation within each cycle.

Proof. Let $\sigma \in S_n$. Form the sequence $1, \sigma(1), \sigma^2(1), \sigma^3(1), \dots$. Since the set $\{1, 2, \dots, n\}$ is finite, for some k we must have $\sigma^k(1)$ equal to a previous term. If $\sigma^k(1) = \sigma^l(1)$ with $l < k$, then $\sigma^{k-l}(1) = 1$. So all terms $\sigma^i(1)$ are distinct until we return to 1, giving the first cycle $(1\ \sigma(1)\ \sigma^2(1)\ \dots\ \sigma^{k-1}(1))$.

Now choose the smallest element not yet in any cycle, say j , and repeat to obtain a cycle $(j\ \sigma(j)\ \sigma^2(j)\ \dots\ \sigma^{l-1}(j))$. Since σ is a bijection, no element of this cycle can appear in a previous cycle.

Repeat until all elements of $\{1, 2, \dots, n\}$ are exhausted. Uniqueness holds because each element j determines the cycle it belongs to: it must be $(j\ \sigma(j)\ \sigma^2(j)\ \dots)$, regardless of which element we start with. $\square$

Definition (Cycle type). Let $\sigma \in S_n$ and write σ in disjoint cycle notation. The *cycle type* of σ is the list of cycle lengths, which is unique up to reordering. By convention, we often omit 1-cycles (fixed points).

Example. $(1\ 2)$ has cycle type 2 (a transposition). $(1\ 2)(3\ 4)$ has cycle type 2, 2 (a double transposition). $(1\ 2\ 3)(4\ 5)$ has cycle type 3, 2.

Lemma (Order of a permutation). Let $\sigma \in S_n$, and write σ as a product of disjoint cycles. Then the order of σ is the least common multiple of the cycle lengths. In particular, a k -cycle has order k .

Proof. Write $\sigma = \tau_1 \tau_2 \dots \tau_l$ where the τ_i are disjoint cycles. Since disjoint cycles commute, $\sigma^m = \tau_1^m \tau_2^m \dots \tau_l^m$ for all $m \geq 1$.

If τ_i has length k_i , then $\tau_i^m = e$ if and only if $k_i \mid m$. Thus $\sigma^m = e$ if and only if $k_i \mid m$ for all i , i.e. m is a common multiple of $k_1, \dots, k_l$. The order of σ

is therefore $\text{lcm}(k_1, \dots, k_l)$. $\square$

Example. Any transposition or double transposition has order 2. The permutation $(1\ 2\ 3)(4\ 5)$ has order $\text{lcm}(3, 2) = 6$.

2.2 Sign of permutations

Remark. While classifying permutations by cycle type is very useful, it is a rather fine division. In this section, we assign a coarser invariant—the *sign*—to each permutation, classifying it as either even or odd.

Proposition (Transposition generation). Every permutation in S_n is a product of transpositions.

Proof. Since every permutation is a product of disjoint cycles, it suffices to show that each cycle is a product of transpositions. Indeed, $(a_1\ a_2\ a_3\ \dots\ a_k) = (a_1\ a_2)(a_2\ a_3)\dots(a_{k-1}\ a_k)$, so a k -cycle is a product of $k - 1$ transpositions. $\square$

Remark. The decomposition into transpositions is not unique. For example,

$$(1\ 2\ 3\ 4\ 5) = (1\ 2)(2\ 3)(3\ 4)(4\ 5) = (1\ 2)(2\ 3)(1\ 2)(3\ 4)(1\ 2)(4\ 5).$$

However, the parity (even or odd) of the number of transpositions is always the same, as the following theorem shows.

Theorem (Parity of transposition decomposition). Let $\sigma \in S_n$. If σ is written as a product of transpositions in two different ways, then the number of transpositions in each decomposition has the same parity (both even or both odd).

Proof. For $\sigma \in S_n$, write $\#(\sigma)$ for the number of cycles in the disjoint cycle notation of σ , including 1-cycles. So $\#(e) = n$ and $\#((1\ 2)) = n - 1$. We claim that for any transposition τ , $\#(\sigma\tau) \equiv \#(\sigma) + 1 \pmod{2}$.

Let $\tau = (c\ d)$ with $c < d$.

- If c and d are in the same cycle of σ , say $(c\ a_2\ \dots\ a_{k-1}\ d\ a_{k+1}\ \dots\ a_{k+l})$, then

$$(c\ a_2\ \dots\ a_{k-1}\ d\ a_{k+1}\ \dots\ a_{k+l})(c\ d) = (c\ a_{k+1}\ \dots\ a_{k+l})(d\ a_2\ \dots\ a_{k-1}),$$

$$\text{so } \#(\sigma\tau) = \#(\sigma) + 1.$$

- If c and d are in different cycles of σ , say $(d\ a_2\ \dots\ a_{k-1})$ and $(c\ a_{k+1}\ \dots\ a_{k+l})$, then

$$(d\ a_2\ \dots\ a_{k-1})(c\ a_{k+1}\ \dots\ a_{k+l})(c\ d) = (c\ a_2\ \dots\ a_{k-1}\ d\ a_{k+1}\ \dots\ a_{k+l}),$$

$$\text{so } \#(\sigma\tau) = \#(\sigma) - 1.$$

In either case, $\#(\sigma\tau) \equiv \#(\sigma) + 1 \pmod{2}$.

Now suppose $\sigma = \tau_1 \cdots \tau_l = \tau'_1 \cdots \tau'_k$. Since the disjoint cycle notation is unique, $\#(\sigma)$ is uniquely determined by σ . Building σ from e by multiplying transpositions one at a time, we get $\#(\sigma) \equiv \#(e) + l \equiv n + l \pmod{2}$ from the first decomposition, and $\#(\sigma) \equiv n + k \pmod{2}$ from the second. Thus $l \equiv k \pmod{2}$. $\square$

Definition (Sign of permutation). Let $\sigma \in S_n$ and write $\sigma = \tau_1 \cdots \tau_l$ as a product of transpositions. The *sign* of σ is $\text{sgn}(\sigma) = (-1)^l$. If $\text{sgn}(\sigma) = 1$, we call σ an *even* permutation; if $\text{sgn}(\sigma) = -1$, we call σ an *odd* permutation.

Remark. The number l depends on the choice of decomposition, but the previous theorem guarantees that its parity does not. Thus $\text{sgn}(\sigma)$ is well-defined.

Theorem (Sign homomorphism). For $n \geq 2$, $\text{sgn}: S_n \rightarrow \{\pm 1\}$ is a surjective group homomorphism.

Proof. Let $\sigma_1 = \tau_1 \cdots \tau_{l_1}$ and $\sigma_2 = \tau'_1 \cdots \tau'_{l_2}$ be products of transpositions. Then $\text{sgn}(\sigma_1\sigma_2) = (-1)^{l_1+l_2} = (-1)^{l_1}(-1)^{l_2} = \text{sgn}(\sigma_1)\text{sgn}(\sigma_2)$, so sgn is a homomorphism.

It is surjective since $\text{sgn}(e) = 1$ and $\text{sgn}((1\ 2)) = -1$. $\square$

Remark. The fact that sgn is a homomorphism is straightforward; the hard part of the theory is showing that sgn is well-defined (i.e. independent of the choice of transposition decomposition).

Lemma (Sign via cycle lengths). A permutation $\sigma \in S_n$ is even if and only if the number of even-length cycles in its disjoint cycle notation is even.

Proof. A k -cycle is a product of $k - 1$ transpositions, so it is even if k is odd and odd if k is even.

Since sgn is a homomorphism, writing σ in disjoint cycle notation as $\sigma = \sigma_1\sigma_2 \cdots \sigma_l$, we get $\text{sgn}(\sigma) = \text{sgn}(\sigma_1) \cdots \text{sgn}(\sigma_l)$. If there are m even-length cycles and p odd-length cycles, then $\text{sgn}(\sigma) = (-1)^m \cdot 1^p = (-1)^m$. Thus $\text{sgn}(\sigma) = 1$ if and only if m is even. $\square$

Remark. Note the potentially confusing convention: odd-length cycles are even permutations, and even-length cycles are odd permutations.

Definition (Alternating group A_n). The *alternating group* A_n is the kernel of $\text{sgn}: S_n \rightarrow \{\pm 1\}$, i.e. A_n is the set of all even permutations in S_n .

Remark. Since A_n is the kernel of a group homomorphism, $A_n \leq S_n$. The sign homomorphism also appears in the definition of the determinant: for an $n \times n$ matrix $A = (a_{ij})$,

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} \cdots a_{n\sigma(n)}.$$

Proposition (Odd permutation proportion). Any subgroup of S_n contains either no odd permutations or exactly half odd permutations.

Proof. Let $H \leq S_n$. If H contains at least one odd permutation τ , then the map $\sigma \mapsto \sigma\tau$ is a bijection from the set of even permutations in H to the set of odd permutations in H (with inverse $\sigma \mapsto \sigma\tau^{-1}$). Thus H contains equally many even and odd permutations. $\square$

3 Lagrange's Theorem

One can model a Rubik's cube with a group, with each possible move corresponding to a group element. Of course, Rubik's cubes of different sizes correspond to different groups.

Suppose I have a $4 \times 4 \times 4$ Rubik's cube, but I want to practice solving a $2 \times 2 \times 2$ Rubik's cube. It is easy. I just have to make sure every time I make a move, I move two layers together. Then I can pretend I am solving a $2 \times 2 \times 2$ cube. This corresponds to picking a particular subgroup of the $4 \times 4 \times 4$ group.

Now what if I have a $3 \times 3 \times 3$ cube? I can still practice solving a $2 \times 2 \times 2$ one. This time, I just look at the corners and pretend that the edges and centers do not exist. Then I am satisfied when the corners are in the right positions, while the centers and edges can be completely scrambled. In this case, we are not taking a subgroup. Instead, we are identifying certain moves together. In particular, we are treating two moves as the same as long as their difference is confined to the centers and edges.

Let G be the $3 \times 3 \times 3$ cube group, and H be the subgroup of G that only permutes the edges and centers. Then for any $a, b \in G$, we think a and b are "the same" if $a^{-1}b \in H$. Then the set of things equivalent to a is $aH = \{ah : h \in H\}$. We call this a *coset*, and the set of cosets form a group.

An immediate question one can ask is: why not $Ha = \{ha : h \in H\}$? In this particular case, the two happen to be the same for all possible a . However, for a general subgroup H , they need not be. We can still define the coset $aH = \{ah : h \in H\}$, but these are less interesting. For example, the set of all $\{aH\}$ will no longer form a group. We will look into these more in-depth in the next chapter. In this chapter, we will first look at results for general cosets. In particular, we will, step by step, prove the things we casually claimed above.

Definition (Cosets). Let $H \leq G$ and $a \in G$. Then the set $aH = \{ah : h \in H\}$ is a *left coset* of H and $Ha = \{ha : h \in H\}$ is a *right coset* of H .

Example.

- (i) Take $2\mathbb{Z} \leq \mathbb{Z}$. Then $6 + 2\mathbb{Z} = \{\text{all even numbers}\} = 0 + 2\mathbb{Z}$.
 $1 + 2\mathbb{Z} = \{\text{all odd numbers}\} = 17 + 2\mathbb{Z}$.

- (ii) Take $G = S_3$, let $H = \langle (1\ 2) \rangle = \{e, (1\ 2)\}$. The left cosets are

$$\begin{aligned} eH &= (1\ 2)H = \{e, (1\ 2)\} \\ (1\ 3)H &= (1\ 2\ 3)H = \{(1\ 3), (1\ 2\ 3)\} \\ (2\ 3)H &= (1\ 3\ 2)H = \{(2\ 3), (1\ 3\ 2)\} \end{aligned}$$

- (iii) Take $G = D_6$ (which is isomorphic to S_3). Recall $D_6 = \langle r, s \mid r^3e = s^2, rs = sr^{-1} \rangle$. Take $H = \langle s \rangle = \{e, s\}$. We have left coset $rH = \{r, rs = sr^{-1}\}$ and the right coset $Hr = \{r, sr\}$. Thus $rH \neq Hr$.

Proposition (Coset equality test). $aH = bH \Leftrightarrow b^{-1}a \in H$.

Proof. $(\Rightarrow)$ Since $a \in aH$, $a \in bH$. Then $a = bh$ for some $h \in H$. So $b^{-1}a = h \in H$.

$(\Leftarrow)$. Let $b^{-1}a = h_0$. Then $a = bh_0$. Then $\forall ah \in aH$, we have $ah = b(h_0h) \in bH$. So $aH \subseteq bH$. Similarly, $bH \subseteq aH$. So $aH = bH$. $\square$

Definition (Partition). Let X be a set, and $X_1, \dots, X_n$ be subsets of X . The X_i are called a *partition* of X if $\bigcup X_i = X$ and $X_i \cap X_j = \emptyset$ for $i \neq j$. i.e. every element is in exactly one of X_i .

Lemma (Coset partition). The left cosets of a subgroup $H \leq G$ partition G , and every coset has the same size.

Proof. For each $a \in G$, $a \in aH$. Thus the union of all cosets gives all of G . Now we have to show that for all $a, b \in G$, the cosets aH and bH are either the same or disjoint.

Suppose that aH and bH are not disjoint. Let $ah_1 = bh_2 \in aH \cap bH$. Then $b^{-1}a = h_2h_1^{-1} \in H$. So $aH = bH$.

To show that they each coset has the same size, note that $f : H \rightarrow aH$ with $f(h) = ah$ is invertible with inverse $f^{-1}(h) = a^{-1}h$. Thus there exists a bijection between them and they have the same size. $\square$

Definition (Index of a subgroup). The *index* of H in G , written $|G : H|$, is the number of left cosets of H in G .

Theorem (Lagrange's theorem). If G is a finite group and H is a subgroup of G , then $|H|$ divides $|G|$. In particular,

$$|H||G : H| = |G|.$$

Note that the converse is not true. If k divides $|G|$, there is not necessarily a subgroup of order k , e.g. $|A_4| = 12$ but there is no subgroup of order 6. However, we will later see that this is true if k is a prime (cf. Cauchy's theorem).

Proof. Suppose that there are $|G : H|$ left cosets in total. Since the left cosets partition G , and each coset has size $|H|$, we have

$$|H||G : H| = |G|. \quad \square$$

Again, the hard part of this proof is to prove that the left cosets partition G and have the same size. If you are asked to prove Lagrange's theorem in exams, that is what you actually have to prove.

Corollary (Element order divisibility). The order of an element divides the order of the group, i.e. for any finite group G and $a \in G$, $\text{ord}(a)$ divides $|G|$.

Proof. Consider the subgroup generated by a , which has order $\text{ord}(a)$. Then by Lagrange's theorem, $\text{ord}(a)$ divides $|G|$. $\square$

Corollary (Exponent divisibility). The exponent of a group divides the order of the group, i.e. for any finite group G and $a \in G$, $a^{|G|} = e$.

Proof. We know that $|G| = k \text{ord}(a)$ for some $k \in \mathbb{N}$. Then $a^{|G|} = (a^{\text{ord}(a)})^k = e^k = e$. $\square$

Corollary (Prime order groups). Groups of prime order are cyclic and are generated by every non-identity element.

Proof. Say $|G| = p$. If $a \in G$ is not the identity, the subgroup generated by a must have order p since it has to divide p . Thus the subgroup generated by a has the same size as G and they must be equal. Then G must be cyclic since it is equal to the subgroup generated by a . $\square$

A useful way to think about cosets is to view them as equivalence classes. To do so, we need to first define what an equivalence class is.

Definition (Equivalence relation). An *equivalence relation* $\sim$ is a relation that is reflexive, symmetric and transitive. i.e.

- (i) $(\forall x) x \sim x$ (reflexivity)
- (ii) $(\forall x, y) x \sim y \Rightarrow y \sim x$ (symmetry)
- (iii) $(\forall x, y, z) [(x \sim y) \wedge (y \sim z) \Rightarrow x \sim z]$ (transitivity)

Example. The following relations are equivalence relations:

- (i) Consider $\mathbb{Z}$. The relation $\equiv_n$ defined as $a \equiv_n b \Leftrightarrow n \mid (a - b)$.
- (ii) Consider the set (formally: class) of all finite groups. Then “is isomorphic to” is an equivalence relation.

Definition (Equivalence class). Given an equivalence relation $\sim$ on A , the *equivalence class* of a is

$$[a]_{\sim} = [a] = \{b \in A : a \sim b\}$$

Proposition (Equivalence class partition). The equivalence classes form a partition of A .

Proof. By reflexivity, we have $a \in [a]$. Thus the equivalence classes cover the whole set. We must now show that for all $a, b \in A$, either $[a] = [b]$ or $[a] \cap [b] = \emptyset$.

Suppose $[a] \cap [b] \neq \emptyset$. Then $\exists c \in [a] \cap [b]$. So $a \sim c, b \sim c$. By symmetry, $c \sim b$.

By transitivity, we have $a \sim b$. Now for all $b' \in [b]$, we have $b \sim b'$. Thus by transitivity, we have $a \sim b'$. Thus $[b] \subseteq [a]$. Similarly, $[a] \subseteq [b]$ and $[a] = [b]$. $\square$

Lemma (Cosets as equivalence classes). Given a group G and a subgroup H , define the equivalence relation on G with $a \sim b$ iff $b^{-1}a \in H$. The equivalence classes are the left cosets of H .

Proof. First show that it is an equivalence relation.

- (i) Reflexivity: Since $aa^{-1} = e \in H$, $a \sim a$.
- (ii) Symmetry: $a \sim b \Rightarrow b^{-1}a \in H \Rightarrow (b^{-1}a)^{-1} = a^{-1}b \in H \Rightarrow b \sim a$.
- (iii) Transitivity: If $a \sim b$ and $b \sim c$, we have $b^{-1}a, c^{-1}b \in H$. So $c^{-1}bb^{-1}a = c^{-1}a \in H$. So $a \sim c$.

To show that the equivalence classes are the cosets, we have $a \sim b \Leftrightarrow b^{-1}a \in H \Leftrightarrow aH = bH$. $\square$

Example. Consider $(\mathbb{Z}, +)$, and for fixed n , take the subgroup $n\mathbb{Z}$. The cosets are $0 + H, 1 + H, \dots, (n-1) + H$. We can write these as $[0], [1], [2] \dots [n]$. To perform arithmetic “mod n ”, define $[a] + [b] = [a+b]$, and $[a][b] = [ab]$. We need to check that it is well-defined, i.e. it doesn't depend on the choice of the representative of $[a]$.
If $[a_1] = [a_2]$ and $[b_1] = [b_2]$, then $a_1 = a_2 + kn$ and $b_1 = b_2 + ln$, then $a_1 + b_1 = a_2 + b_2 + n(k+l)$ and $a_1b_1 = a_2b_2 + n(kb_2 + la_2 + kln)$. So $[a_1 + b_1] = [a_2 + b_2]$ and $[a_1b_1] = [a_2b_2]$.

We have seen that $(\mathbb{Z}_n, +_n)$ is a group. What happens with multiplication? We can only take elements which have inverses (these are called units, cf. IB Groups, Rings and Modules). Call the set of them $U_n = \{[a] : (a, n) = 1\}$. We'll see these are the units.

Definition (Euler totient function). (Euler totient function) $\phi(n) = |U_n|$.

Example. If p is a prime, $\phi(p) = p - 1$. $\phi(4) = 2$.

Proposition (Group of units mod n). U_n is a group under multiplication mod n .

Proof. The operation is well-defined as shown above. To check the axioms:

- 0. Closure: if a, b are coprime to n , then $a \cdot b$ is also coprime to n . So $[a], [b] \in U_n \Rightarrow [a] \cdot [b] = [a \cdot b] \in U_n$
- 1. Identity: $[1]$
- 2. Let $[a] \in U_n$. Consider the map $U_n \rightarrow U_n$ with $[c] \mapsto [ac]$. This is injective: if $[ac_1] = [ac_2]$, then n divides $a(c_1 - c_2)$. Since a is coprime to n , n divides

$c_1 - c_2$, so $[c_1] = [c_2]$. Since U_n is finite, any injection ($U_n \rightarrow U_n$) is also a surjection. So there exists a c such that $[ac] = [a][c] = 1$. So $[c] = [a]^{-1}$.

3. Associativity (and also commutativity): inherited from $\mathbb{Z}$. $\square$

Theorem (Fermat-Euler theorem). Let $n \in \mathbb{N}$ and $a \in \mathbb{Z}$ coprime to n . Then

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

In particular, (Fermat's Little Theorem) if $n = p$ is a prime, then for any a not a multiple of p .

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. As a is coprime with n , $[a] \in U_n$. Then $[a]^{|U_n|} = [1]$, i.e. $a^{\phi(n)} \equiv 1 \pmod{n}$. $\square$

3.1 Small groups

Remark. Lagrange's theorem severely restricts the possible structures of small groups. We now use it to enumerate the subgroups of certain familiar groups and to classify all groups of orders 4 and 6.

Example (Subgroups of D_{10}). Recall that $D_{10} = \langle r, s \mid r^5 = s^2 = e, srs^{-1} = r^{-1} \rangle$, with elements $e, r, r^2, r^3, r^4, s, rs, r^2s, r^3s, r^4s$. By Lagrange's theorem, any subgroup of D_{10} must have order dividing $|D_{10}| = 10$, so the possible orders are 1, 2, 5 and 10.

1: $\{e\}$.

2: Each of the 5 reflections s, rs, r^2s, r^3s, r^4s has order 2, giving 5 distinct subgroups of order 2.

5: Any subgroup of order 5 is cyclic, since 5 is prime. Such a subgroup is generated by an element of order 5, i.e. any of r, r^2, r^3, r^4 . These all generate the same subgroup $\langle r \rangle$.

10: D_{10} itself.

Example (Subgroups of D_8). Recall that $D_8 = \langle r, s \mid r^4 = s^2 = e, srs^{-1} = r^{-1} \rangle$. By Lagrange's theorem, any subgroup must have order dividing $|D_8| = 8$, so the possible orders are 1, 2, 4 and 8.

1: $\{e\}$.

2: There are 5 elements of order 2: the four reflections s, rs, r^2s, r^3s and the rotation r^2 . Each generates a distinct subgroup of order 2.

4: There is one cyclic subgroup of order 4, namely $\langle r \rangle \cong C_4$. There are also two non-cyclic subgroups of order 4, both isomorphic to

$C_2 \times C_2$: $\{e, r^2, s, r^2s\}$ and $\{e, r^2, rs, r^3s\}$.

8: D_8 itself.

Proposition (Classification of groups of order 4). Any group of order 4 is isomorphic to either C_4 or $C_2 \times C_2$.

Proof. Let G be a group with $|G| = 4$. By Lagrange's theorem, the possible orders of elements of G are 1, 2 and 4.

If G contains an element a of order 4, then $G = \langle a \rangle \cong C_4$.

Otherwise, every non-identity element has order 2. We first show that G is abelian: for any $a, b \in G$, we have $(ab)^2 = e$, so $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$, where the last equality uses $a^2 = b^2 = e$.

Now pick two distinct non-identity elements $b, c \in G$. Then $\langle b \rangle = \{e, b\}$ and $\langle c \rangle = \{e, c\}$ are subgroups with $\langle b \rangle \cap \langle c \rangle = \{e\}$. Since G is abelian, elements of $\langle b \rangle$ and $\langle c \rangle$ commute. The product bc is the unique remaining element of G (it is distinct from e, b and c), so every element of G can be written as a product of elements of $\langle b \rangle$ and $\langle c \rangle$. By the direct product theorem, $G \cong \langle b \rangle \times \langle c \rangle \cong C_2 \times C_2$. $\square$

Proposition (Classification of groups of order 6). Any group of order 6 is isomorphic to either C_6 or $D_6 \cong S_3$.

Remark. The proof of this classification uses quotient groups and is deferred to the next section.

3.2 Left and right cosets

Remark. We have seen that $|aH| = |H| = |Ha|$ for any subgroup $H \leq G$ and any $a \in G$, so left and right cosets always have the same size. However, the sets aH and Ha need not be equal.

Proposition (Cosets in abelian groups). If G is an abelian group, then $aH = Ha$ for all $a \in G$ and all $H \leq G$.

Proof. For any $ah \in aH$, commutativity gives $ah = ha \in Ha$, so $aH \subseteq Ha$. The reverse inclusion is identical, so $aH = Ha$. $\square$

Example. Let $G = (\mathbb{Z}, +)$ and $H = 2\mathbb{Z}$. Then $0 + 2\mathbb{Z} = 2\mathbb{Z} + 0$ is the set of even integers, and $1 + 2\mathbb{Z} = 2\mathbb{Z} + 1$ is the set of odd integers.

Example (Left cosets equal right cosets in a non-abelian group). Let $G = D_6 = \langle r, s \mid r^3 = s^2 = e, srs^{-1} = r^{-1} \rangle$ and let $U = \langle r \rangle = \{e, r, r^2\}$.

Since $|G : U| = 2$, the cosets of U partition G into exactly two classes: U itself and $sU = \{s, sr, sr^2\} = \{s, r^2s, rs\} = Us$. Thus $aU = Ua$ for all $a \in G$, even though G is not abelian.

Example (Left cosets not equal to right cosets). Let $G = D_6$ as above and let $H = \langle s \rangle = \{e, s\}$. The left cosets are

$$H = \{e, s\}, \quad rH = \{r, rs\}, \quad r^2H = \{r^2, r^2s\},$$

while the right cosets are

$$H = \{e, s\}, \quad Hr = \{r, sr\} = \{r, r^2s\}, \quad Hr^2 = \{r^2, sr^2\} = \{r^2, rs\}.$$

In particular, $rH = \{r, rs\} \neq \{r, r^2s\} = Hr$, so the left and right cosets do not coincide.

Remark. The distinction between left and right cosets will play a central role in the next section, where we identify the subgroups for which $aH = Ha$ always holds.

4 Quotient groups

Remark. In the previous section, when attempting to pretend that a $3 \times 3 \times 3$ Rubik's cube is a $2 \times 2 \times 2$ one, we formed the cosets aH and claimed that they form a group. We also noted that this is not the case for an arbitrary subgroup H , but only for subgroups satisfying $aH = Ha$ for all $a \in G$. We now study these special subgroups.

4.1 Normal subgroups

Definition (Normal subgroup). A subgroup K of a group G is a *normal subgroup*, written $K \triangleleft G$, if

$$(\forall a \in G)(\forall k \in K) aka^{-1} \in K.$$

The following conditions are equivalent to normality:

- (i) $(\forall a \in G) aK = Ka$ (every left coset equals the corresponding right coset);
- (ii) $(\forall a \in G) aKa^{-1} = K$ (the subgroup is invariant under conjugation).

Example (Normal and non-normal subgroups of D_6). Let $G = D_6 = \langle r, s \mid r^3 = s^2 = e, srs^{-1} = r^{-1} \rangle$. We showed in the previous subsection that $a\langle r \rangle = \langle r \rangle a$ for all $a \in G$, so $\langle r \rangle \triangleleft D_6$. On the other hand, $r\langle s \rangle \neq \langle s \rangle r$, so $\langle s \rangle$ is not a normal subgroup of D_6 .

Remark. Every group G has at least two normal subgroups: the trivial subgroup $\{e\} \triangleleft G$ and $G \triangleleft G$.

Lemma (Normality criteria).

- (i) Every subgroup of index 2 is normal.
- (ii) Every subgroup of an abelian group is normal.

Proof.

- (i) If $K \leq G$ has index 2, then there are exactly two left cosets (K and $G \setminus K$) and two right cosets (K and $G \setminus K$). Since $eK = Ke = K$, the other left and right coset must both equal $G \setminus K$. Thus $aK = Ka$ for all $a \in G$.
- (ii) For all $a \in G$ and $k \in K$, commutativity gives $aka^{-1} = aa^{-1}k = k \in K$. $\square$

Proposition (Kernel normality). Let $f: G \rightarrow H$ be a group homomorphism. Then $\ker f \triangleleft G$.

Proof. For any $a \in G$ and $k \in \ker f$, we have

$$f(aka^{-1}) = f(a)f(k)f(a)^{-1} = f(a)e_H f(a)^{-1} = e_H.$$

Thus $aka^{-1} \in \ker f$, so $\ker f \triangleleft G$. $\square$

Remark. We will see in the next subsection that, conversely, every normal subgroup is the kernel of some homomorphism.

Example (Normal subgroup of D_8). Consider $G = D_8 = \langle r, s \mid r^4 = s^2 = e, srs^{-1} = r^{-1} \rangle$ and let $K = \langle r^2 \rangle = \{e, r^2\}$. We verify that $K \triangleleft G$. Every element of G is of the form r^ℓ or sr^ℓ for some ℓ . For $a = r^\ell$, we have $r^\ell r^2 r^{-\ell} = r^2 \in K$. For $a = sr^\ell$, we have

$$sr^\ell \cdot r^2 \cdot (sr^\ell)^{-1} = sr^2 s^{-1} = (srs^{-1})^2 = (r^{-1})^2 = r^{-2} = r^2 \in K.$$

Thus $aka^{-1} \in K$ for all $a \in G$ and $k \in K$, so $K \triangleleft G$.

Proposition (Classification of groups of order 6). Any group of order 6 is isomorphic to either C_6 or $D_6 \cong S_3$.

Proof. Let G be a group with $|G| = 6$. By Lagrange's theorem, the possible orders of elements are 1, 2, 3 and 6.

If G contains an element of order 6, then G is cyclic, so $G \cong C_6$.

Otherwise, every non-identity element has order 2 or 3. Not all non-identity elements can have order 2: if they did, then G would be abelian (as shown in the proof of the classification of groups of order 4), and for any two distinct non-identity elements a, b , the set $\{e, a, b, ab\}$ would be a subgroup of order 4, contradicting Lagrange's theorem since $4 \nmid 6$. So there exists an element $r \in G$ of order 3.

The subgroup $\langle r \rangle$ has index $|G : \langle r \rangle| = 2$, so $\langle r \rangle \triangleleft G$ by the normality criteria lemma. Furthermore, not all non-identity elements can have order 3, since elements of order 3 come in inverse pairs $\{g, g^{-1}\}$ (note $g \neq g^{-1}$ when $\text{ord}(g) = 3$), contributing an even number of such elements, but $|G| - 1 = 5$ is odd. So there also exists an element $s \in G$ of order 2.

Since $\langle r \rangle$ is normal, $srs^{-1} \in \langle r \rangle = \{e, r, r^2\}$. If $srs^{-1} = e$, then $r = e$, a contradiction. If $srs^{-1} = r$, then $sr = rs$, and since $\langle s \rangle \cap \langle r \rangle = \{e\}$ (as $\text{ord}(s) = 2$ and $\text{ord}(r) = 3$), the element sr has order $\text{lcm}(2, 3) = 6$, contradicting our assumption. Therefore $srs^{-1} = r^2 = r^{-1}$, so $G = \langle r, s \mid r^3 = s^2 = e, srs^{-1} = r^{-1} \rangle \cong D_6$. $\square$

4.2 Quotient groups

Proposition (Quotient group construction). Let $K \triangleleft G$. Then the set of left cosets of K in G forms a group under the operation $aK \cdot bK = (ab)K$.

Proof. We first show that the operation is well-defined. Suppose $aK = a'K$ and $bK = b'K$; we must show that $(ab)K = (a'b')K$. We have $a' = ak_1$ and $b' = bk_2$ for some $k_1, k_2 \in K$. Since $K \triangleleft G$, we have $b^{-1}k_1b \in K$; call it k_3 . Then $k_1b = bk_3$, so $a'b' = ak_1bk_2 = abk_3k_2 \in (ab)K$. Thus the operation does not depend on the choice of coset representatives.

1. Closure: if aK and bK are cosets, then $(ab)K$ is also a coset.
2. Identity: the identity is $eK = K$.
3. Inverses: the inverse of aK is $a^{-1}K$.
4. Associativity: follows from the associativity of G . □

Definition (Quotient group). Given a group G and a normal subgroup K , the *quotient group* (or *factor group*) of G by K , written G/K , is the set of left cosets of K in G under the operation $aK \cdot bK = (ab)K$.

Remark. The set of left cosets exists for any subgroup $H \leq G$, but the coset multiplication above is well-defined only when H is normal.

Example (Quotient groups).

- (i) Let $G = \mathbb{Z}$ and $K = n\mathbb{Z}$ (which is normal since $\mathbb{Z}$ is abelian). The cosets are $k + n\mathbb{Z}$ for $0 \leq k < n$, so $\mathbb{Z}/(n\mathbb{Z}) \cong \mathbb{Z}_n$. Since $n\mathbb{Z}$ are the only subgroups of $\mathbb{Z}$, these are the only quotient groups of $\mathbb{Z}$.
- (ii) Let $K = \langle r \rangle \triangleleft D_6$. There are two cosets, K and sK , so D_6/K has order 2 and $D_6/K \cong C_2$.
- (iii) Let $K = \langle r^2 \rangle \triangleleft D_8$. Then $|G/K| = |G|/|K| = 8/2 = 4$, and the four cosets are $G/K = \{K, rK = r^3K, sK = sr^2K, srK = sr^3K\}$. Every non-identity element of G/K has order 2, so $G/K \cong C_2 \times C_2$.

Remark. If G is abelian, then G/K is also abelian for any $K \triangleleft G$.

Remark. Quotient groups are *not* subgroups of G : their elements are cosets, not elements of G . For instance, $\mathbb{Z}/(n\mathbb{Z}) \cong C_n$ is finite, even though every subgroup of $\mathbb{Z}$ is infinite.

Example (Non-example: failure of well-definedness). Consider D_6 with $H = \langle s \rangle$, which is not a normal subgroup. We have $rH \cdot r^2H = r^3H = H$,

but $rH = rsH$ and $r^2H = srH$ (by examining the individual elements), so $rsH \cdot srH = r^2H \neq H$. The coset multiplication depends on the choice of representatives, and is therefore not well-defined.

Lemma (Quotient map). Let $K \triangleleft G$. The *quotient map* $q: G \rightarrow G/K$ defined by $g \mapsto gK$ is a surjective group homomorphism.

Proof. For any $a, b \in G$, we have $q(ab) = (ab)K = (aK)(bK) = q(a)q(b)$, so q is a homomorphism. Surjectivity is immediate: for any coset $aK \in G/K$, we have $q(a) = aK$. $\square$

Remark. The kernel of the quotient map is K itself, since $q(g) = K$ if and only if $gK = K$ if and only if $g \in K$. This shows that every normal subgroup is the kernel of some homomorphism, establishing the converse of the kernel normality proposition.

Proposition (Quotient of cyclic groups). Every quotient of a cyclic group is cyclic.

Proof. Let $G = C_n = \langle c \rangle$ and let $H \leq C_n$. Since every subgroup of a cyclic group is cyclic, we have $H = \langle c^k \rangle \cong C_\ell$ for some k, ℓ with $k\ell = n$. Then $C_n/H = \{H, cH, c^2H, \dots, c^{k-1}H\} = \langle cH \rangle \cong C_k$. $\square$

4.3 The Isomorphism Theorem

Remark. We now come to the central theorem of this chapter.

Theorem (The Isomorphism Theorem). Let $f: G \rightarrow H$ be a group homomorphism with kernel K . Then $K \triangleleft G$ and $G/K \cong \text{im } f$.

Proof. We have already proved that $K \triangleleft G$. Define $\theta: G/K \rightarrow \text{im } f$ by $\theta(aK) = f(a)$.

Well-defined: If $a_1K = a_2K$, then $a_2^{-1}a_1 \in K$, so

$$f(a_2)^{-1}f(a_1) = f(a_2^{-1}a_1) = e.$$

Thus $f(a_1) = f(a_2)$, and hence $\theta(a_1K) = \theta(a_2K)$.

Homomorphism:

$$\theta(aK \cdot bK) = \theta(abK) = f(ab) = f(a)f(b) = \theta(aK)\theta(bK).$$

Injective: Suppose $\theta(aK) = \theta(bK)$. Then $f(a) = f(b)$, so $f(b)^{-1}f(a) = e$, hence $b^{-1}a \in K$, and therefore $aK = bK$.

Surjective: For any $h \in \text{im } f$, there exists $a \in G$ with $f(a) = h$, so $\theta(aK) = h$. Thus θ is an isomorphism $G/K \cong \text{im } f$. $\square$

Remark.

- If f is injective, then $K = \{e\}$, so $G/\{e\} \cong G$ and G is isomorphic to a subgroup of H . We can think of f as an inclusion map.
- If f is surjective, then $\text{im } f = H$, so $G/K \cong H$.

Example (Applications of the Isomorphism Theorem).

- (i) Consider $f: \text{GL}_n(\mathbb{R}) \rightarrow \mathbb{R}^*$ defined by $A \mapsto \det A$. Then $\ker f = \text{SL}_n(\mathbb{R})$, and f is surjective since for all $\lambda \in \mathbb{R}^*$,

$$\det \begin{pmatrix} \lambda & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 1 \end{pmatrix} = \lambda.$$

By the Isomorphism Theorem, $\text{GL}_n(\mathbb{R})/\text{SL}_n(\mathbb{R}) \cong \mathbb{R}^*$.

- (ii) Define $\theta: (\mathbb{R}, +) \rightarrow (\mathbb{C}^*, \times)$ by $r \mapsto \exp(2\pi ir)$. This is a homomorphism since $\theta(r+s) = \exp(2\pi i(r+s)) = \exp(2\pi ir)\exp(2\pi is) = \theta(r)\theta(s)$. The kernel is $\mathbb{Z} \triangleleft \mathbb{R}$ and the image is the unit circle $(S^1, \times)$, so $\mathbb{R}/\mathbb{Z} \cong (S^1, \times)$.
- (iii) Let $G = (\mathbb{Z}_p^*, \times)$ for an odd prime p , and define $f: G \rightarrow G$ by $a \mapsto a^2$. This is a homomorphism since G is abelian: $(ab)^2 = a^2b^2$. The kernel is $\{\pm 1\} = \{1, p-1\}$, so $|\text{im } f| = |G/\ker f| = \frac{p-1}{2}$. The elements of $\text{im } f$ are called *quadratic residues modulo p* .

Lemma (Classification of cyclic groups). Every cyclic group is isomorphic to either $\mathbb{Z}$ or $\mathbb{Z}/(n\mathbb{Z})$ for some $n \in \mathbb{N}$.

Proof. Let $G = \langle c \rangle$. Define $f: \mathbb{Z} \rightarrow G$ by $m \mapsto c^m$. This is a homomorphism since $c^{m_1+m_2} = c^{m_1}c^{m_2}$, and it is surjective since $G = \{c^m : m \in \mathbb{Z}\}$ by definition. Since the only subgroups of $\mathbb{Z}$ are $n\mathbb{Z}$ for $n \geq 0$, there are three cases:

- (i) $\ker f = \{0\}$, so f is an isomorphism and $G \cong \mathbb{Z}$;
- (ii) $\ker f = \mathbb{Z}$, so $G \cong \mathbb{Z}/\mathbb{Z} = \{e\} \cong C_1$;
- (iii) $\ker f = n\mathbb{Z}$ for some $n \geq 2$, so $G \cong \mathbb{Z}/(n\mathbb{Z})$. □

Definition (Simple group). A group G is *simple* if its only normal subgroups are $\{e\}$ and G itself.

Example (Simple groups). The cyclic group C_p is simple for any prime p , since C_p has no proper subgroups at all, let alone normal ones. The alternating group A_5 is also simple, which we will prove after Chapter 6.

Remark. The finite simple groups are the building blocks of all finite groups. If $K \triangleleft G$ with $K \neq \{e\}$ and $K \neq G$, we can form the quotient G/K ; if G/K is not simple, we repeat. In this way, every finite group can be decomposed into simple “factors.” All finite simple groups have been classified (see *The Atlas of Finite Groups*).

5 Group actions

Remark. Recall that we introduced groups to model symmetries and permutations. Intuitively, elements of groups are supposed to “do things.” However, as we developed group theory, we abstracted these away and focused on how elements combine to form new elements. *Group actions* recapture this idea and make each group element correspond to some function.

5.1 Group acting on sets

Definition (Group action). Let X be a set and G be a group. An *action* of G on X is a homomorphism $\varphi: G \rightarrow \text{Sym } X$.

Remark. The homomorphism φ turns each element $g \in G$ into a permutation of X , in a way that respects the group structure. Instead of writing $\varphi(g)(x)$, we usually write $g(x)$ or gx .

Proposition (Action axioms, equivalent formulation). Let X be a set and G be a group. Then $\varphi: G \rightarrow \text{Sym } X$ is a homomorphism (i.e. an action) if and only if $\theta: G \times X \rightarrow X$ defined by $\theta(g, x) = \varphi(g)(x)$ satisfies

0. $(\forall g \in G)(\forall x \in X) \theta(g, x) \in X$.
1. $(\forall x \in X) \theta(e, x) = x$.
2. $(\forall g, h \in G)(\forall x \in X) \theta(g, \theta(h, x)) = \theta(gh, x)$.

Remark. This criterion is *almost* the definition of a homomorphism. Here we do not explicitly require $\theta(g, \cdot)$ to be a bijection, but instead require $\theta(e, \cdot)$ to be the identity function. This automatically ensures that $\theta(g, \cdot)$ is a bijection: composing with $\theta(g^{-1}, \cdot)$ gives $\theta(e, \cdot)$, which is the identity, so $\theta(g, \cdot)$ has an inverse. This is usually easier to verify in practice.

Example (Group actions).

- (i) *Trivial action:* For any group G acting on any set X , we can define $\varphi(g) = 1_X$ for all g , i.e. G does nothing.
- (ii) S_n acts on $\{1, \dots, n\}$ by permutation.
- (iii) D_{2n} acts on the vertices of a regular n -gon (or the set $\{1, \dots, n\}$).
- (iv) The rotations of a cube act on the faces, vertices, diagonals, or axes of the cube.

Remark. Different groups can act on the same set, and the same group can act on different sets.

Definition (Kernel of action). The *kernel* of an action of G on X is $\ker \varphi = \{g \in G : \varphi(g) = 1_X\}$.

Remark. By the Isomorphism Theorem, $\ker \varphi \triangleleft G$ and $G/\ker \varphi$ is isomorphic to a subgroup of $\text{Sym } X$.

Example (Kernels of actions).

- (i) D_{2n} acting on $\{1, 2, \dots, n\}$ gives $\varphi: D_{2n} \rightarrow S_n$ with kernel $\{e\}$.
- (ii) Let G be the rotations of a cube, acting on the three axes x, y, z through the faces. This gives $\varphi: G \rightarrow S_3$. Any rotation by 180° fixes every axis, so the kernel contains at least 4 elements: e and the three 180° rotations. We will see later that these are exactly the kernel.

Definition (Faithful action). An action is *faithful* if its kernel is $\{e\}$.

5.2 Orbits and Stabilizers

Definition (Orbit). Given an action of G on X , the *orbit* of an element $x \in X$ is

$$\text{orb}(x) = G(x) = \{y \in X : (\exists g \in G) g(x) = y\}.$$

Intuitively, this is the set of elements that x can be mapped to.

Definition (Stabilizer). The *stabilizer* of x is

$$\text{stab}(x) = G_x = \{g \in G : g(x) = x\} \subseteq G.$$

Intuitively, this is the set of elements in G that fix x .

Lemma (Stabilizer subgroup). $\text{stab}(x)$ is a subgroup of G .

Proof. Since $e(x) = x$, we have $e \in \text{stab}(x)$, so $\text{stab}(x)$ is non-empty. If $g, h \in \text{stab}(x)$, then $gh^{-1}(x) = g(h^{-1}(x)) = g(x) = x$, so $gh^{-1} \in \text{stab}(x)$. Hence $\text{stab}(x)$ is a subgroup. $\square$

Example (Orbits and stabilizers).

- (i) Consider D_8 acting on the corners of the square $X = \{1, 2, 3, 4\}$. Then $\text{orb}(1) = X$ since 1 can be sent anywhere by rotations, and $\text{stab}(1) = \{e, s\}$ where s is the reflection in the line through vertex 1.
- (ii) Consider the rotations of a cube acting on the three axes x, y, z . Then $\text{orb}(x) = \{x, y, z\}$, and $\text{stab}(x)$ consists of e , the 180° rotations, and the rotations about the x -axis.

Definition (Transitive action). An action of G on X is *transitive* if $\text{orb}(x) = X$ for all $x \in X$, i.e. any element of X can be reached from any other.

Lemma (Orbit partition). The orbits of an action partition X .

Proof. Every $x \in X$ lies in $\text{orb}(x)$, since $e(x) = x$. So the orbits cover X .

Now suppose $z \in \text{orb}(x) \cap \text{orb}(y)$. We show that $\text{orb}(x) = \text{orb}(y)$. Write $z = g_1(x)$ and $z = g_2(y)$ for some $g_1, g_2 \in G$. Then $y = g_2^{-1}g_1(x)$.

For any $w = g_3(y) \in \text{orb}(y)$, we have $w = g_3g_2^{-1}g_1(x) \in \text{orb}(x)$. Thus $\text{orb}(y) \subseteq \text{orb}(x)$, and by symmetry $\text{orb}(x) \subseteq \text{orb}(y)$. $\square$

Remark. Suppose G acts on X and fix $x \in X$. Each $g \in G$ gives an element $g(x) \in \text{orb}(x)$, and every element of $\text{orb}(x)$ arises this way. However, different elements of G can give the same element of $\text{orb}(x)$: if $g \in \text{stab}(x)$, then hg and h give the same element, since $hg(x) = h(g(x)) = h(x)$. So there is a correspondence between elements of $\text{orb}(x)$ and elements of G “up to $\text{stab}(x)$.”

Theorem (Orbit-stabilizer theorem). Let the group G act on X . Then there is a bijection between $\text{orb}(x)$ and cosets of $\text{stab}(x)$ in G . In particular, if G is finite, then

$$|\text{orb}(x)| \cdot |\text{stab}(x)| = |G|.$$

Proof. Define

$$\begin{aligned} \theta: (G : \text{stab}(x)) &\rightarrow \text{orb}(x) \\ g \text{ stab}(x) &\mapsto g(x). \end{aligned}$$

Well-defined: If $g \text{ stab}(x) = h \text{ stab}(x)$, then $h = gk$ for some $k \in \text{stab}(x)$, so $h(x) = g(k(x)) = g(x)$.

Surjective: For any $y \in \text{orb}(x)$, there exists $g \in G$ with $g(x) = y$, so $\theta(g \text{ stab}(x)) = y$.

Injective: If $g(x) = h(x)$, then $h^{-1}g(x) = x$, so $h^{-1}g \in \text{stab}(x)$, hence $g \text{ stab}(x) = h \text{ stab}(x)$.

Thus $|\text{orb}(x)|$ equals the number of cosets of $\text{stab}(x)$, and the result follows from Lagrange's theorem. $\square$

Remark. An important application of the orbit-stabilizer theorem is determining group sizes. To find the order of the symmetry group of, say, a pyramid, we find something for it to act on, pick a convenient element, and compute the orbit and stabilizer sizes.

Example (Orbit-stabilizer calculations).

- (i) D_{2n} acts on the vertices $\{1, 2, \dots, n\}$ transitively, so $|\text{orb}(1)| = n$. Also $\text{stab}(1) = \{e, s\}$ where s is the reflection through vertex 1. So $|D_{2n}| = |\text{orb}(1)| \cdot |\text{stab}(1)| = 2n$.
- (ii) Let $\langle(1\ 2)\rangle$ act on $\{1, 2, 3\}$. Then $\text{orb}(1) = \{1, 2\}$ and $\text{stab}(1) = \{e\}$, while $\text{orb}(3) = \{3\}$ and $\text{stab}(3) = \langle(1\ 2)\rangle$.
- (iii) Consider S_4 acting on $\{1, 2, 3, 4\}$. Since the action is transitive, $|\text{orb}(1)| = 4$, so $|\text{stab}(1)| = 24/4 = 6$. The subgroup $S_{\{2,3,4\}} \cong S_3$ fixes 1, so $S_{\{2,3,4\}} \leq \text{stab}(1)$. Since $|S_3| = 6 = |\text{stab}(1)|$, we conclude $\text{stab}(1) = S_{\{2,3,4\}}$.

Remark. If the action is transitive, then every orbit has size $|X|$ and hence all stabilizers have the same size $|G|/|X|$.

5.3 Important actions

Remark. Given any group G , there are several important actions we can define. In particular, we will define the *conjugation* action, which is a central concept in group theory: the whole of the next chapter is devoted to studying conjugation in the symmetric groups. We begin with two simpler examples.

Lemma (Left regular action). Any group G acts on itself by left multiplication. This action is faithful and transitive.

Proof. We verify the action axioms:

1. $(\forall g \in G)(\forall x \in G) g(x) = gx \in G$ by closure.
2. $(\forall x \in G) ex = x$ by definition of identity.
3. $g(hx) = (gh)x$ by associativity.

Faithful: If $gx = x$ for all $x \in G$, then taking $x = e$ gives $g = e$.

Transitive: For any $x, y \in G$, we have $(yx^{-1})(x) = y$. $\square$

Theorem (Cayley's theorem). Every group is isomorphic to a subgroup of some symmetric group.

Proof. The left regular action gives a homomorphism $\varphi: G \rightarrow \text{Sym } G$ with $\ker \varphi = \{e\}$, since the action is faithful. By the Isomorphism Theorem, $G \cong \text{im } \varphi \leq \text{Sym } G$. $\square$

Lemma (Left coset action). Let $H \leq G$. Then G acts on the set of left cosets of H by left multiplication, and this action is transitive.

Proof. We verify the action axioms:

0. $g(aH) = (ga)H$ is a coset of H .
1. $e(aH) = (ea)H = aH$.
2. $g_1(g_2(aH)) = g_1((g_2a)H) = (g_1g_2a)H = (g_1g_2)(aH)$.

Transitive: Given aH, bH , we have $(ba^{-1})(aH) = bH$. $\square$

Remark. When $H = \{e\}$, the left coset action reduces to the left regular action, since $G/\{e\} \cong G$.

Definition (Conjugation of element). The *conjugation* of $a \in G$ by $b \in G$ is the element $bab^{-1} \in G$. If there exists $b \in G$ such that $c = bab^{-1}$, we say a and c are *conjugate*.

Remark. The expression bab^{-1} is familiar from linear algebra: if b is a change-of-basis matrix and a is a linear map, then bab^{-1} is the same map viewed in the new basis. In general, conjugate elements are “the same” in an appropriate sense. For example, we will later show that in S_n , two elements are conjugate if and only if they have the same cycle type. Conjugate elements share many properties, such as their order.

Lemma (Conjugation action). Any group G acts on itself by conjugation, i.e. $g(x) = gxg^{-1}$.

Proof. We verify the action axioms:

0. $g(x) = gxg^{-1} \in G$ for all $g, x \in G$.
1. $e(x) = exe^{-1} = x$.
2. $g(h(x)) = g(hxh^{-1}) = ghxh^{-1}g^{-1} = (gh)x(gh)^{-1} = (gh)(x)$. $\square$

Definition (Conjugacy class and centralizer). The *conjugacy class* of

$a \in G$ is the orbit of a under the conjugation action:

$$\text{ccl}(a) = \{b \in G : (\exists g \in G) gag^{-1} = b\}.$$

The *centralizer* of a is the stabilizer of a under conjugation:

$$C_G(a) = \{g \in G : gag^{-1} = a\} = \{g \in G : ga = ag\}.$$

Remark. The centralizer $C_G(a)$ consists of all elements that commute with a . We can also consider the elements that commute with *every* element.

Definition (Center of group). The *center* of G is

$$Z(G) = \{g \in G : (\forall a \in G) ga = ag\} = \bigcap_{a \in G} C_G(a).$$

It is sometimes written $C(G)$ instead of $Z(G)$.

Lemma (Conjugation action on normal subgroups). Let $K \triangleleft G$. Then G acts by conjugation on K .

Proof. The only axiom that requires checking is closure: by definition of a normal subgroup, $gkg^{-1} \in K$ for all $g \in G$ and $k \in K$. The remaining axioms follow from the conjugation action on G . $\square$

Proposition (Normal subgroups and conjugacy classes). A subgroup is normal if and only if it is a union of conjugacy classes.

Proof. Let $K \triangleleft G$. For any $k \in K$ and $g \in G$, we have $gkg^{-1} \in K$, so $\text{ccl}(k) \subseteq K$. Hence K is a union of conjugacy classes.

Conversely, suppose $K \leq G$ is a union of conjugacy classes. Then for all $k \in K$ and $g \in G$, we have $gkg^{-1} \in \text{ccl}(k) \subseteq K$, so $K \triangleleft G$. $\square$

Lemma (Conjugation action on subgroups). Let X be the set of subgroups of G . Then G acts on X by conjugation: $g(H) = gHg^{-1}$.

Proof. We verify the action axioms:

0. If $H \leq G$, then gHg^{-1} is a subgroup: $e = geg^{-1} \in gHg^{-1}$, and for $gag^{-1}, gbg^{-1} \in gHg^{-1}$, we have $(gag^{-1})(gbg^{-1})^{-1} = g(ab^{-1})g^{-1} \in gHg^{-1}$.
1. $eHe^{-1} = H$.
2. $g_1(g_2Hg_2^{-1})g_1^{-1} = (g_1g_2)H(g_1g_2)^{-1}$. $\square$

Remark. Under this action, the normal subgroups of G are precisely the subgroups with singleton orbits (i.e. $gKg^{-1} = K$ for all g).

Definition (Normalizer). The *normalizer* of a subgroup $H \leq G$ is the stabilizer of H under the conjugation action on subgroups:

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

Remark. We always have $H \leq N_G(H)$, and $N_G(H)$ is the largest subgroup of G in which H is normal, hence the name.

Lemma (Conjugate stabilizers). Let G act on X and let $g \in G$, $x \in X$. Then $\text{stab}(g(x)) = g \text{stab}(x) g^{-1}$. In particular, stabilizers of elements in the same orbit are conjugate subgroups.

5.4 Applications

Example (Rotations of a cube). Let G^+ be the group of rotations of a cube, acting on the set of vertices X . Then $|X| = 8$. Since the action is transitive, the orbit of any vertex is all of X . The stabilizer of vertex 1 consists of the rotations about the diagonal through 1 and the opposite vertex, of which there are 3. So $|G^+| = |\text{orb}(1)| \cdot |\text{stab}(1)| = 8 \cdot 3 = 24$.

Proposition (Order bound for finite simple groups). Let G be a finite simple group of order greater than 2, and let $H \leq G$ have index $n \neq 1$. Then $|G| \leq n!/2$.

Proof. Consider the left coset action of G on the cosets of H . This gives a homomorphism $\varphi: G \rightarrow S_n$. Since $H \neq G$, the map φ is non-trivial, so $\ker \varphi \neq G$. Since $\ker \varphi \triangleleft G$ and G is simple, we must have $\ker \varphi = \{e\}$. By the Isomorphism Theorem, $G \cong \text{im } \varphi \leq S_n$, so $|G| \leq n!$.

We refine this using $\text{sgn} \circ \varphi: G \rightarrow \{\pm 1\}$. Let $K = \ker(\text{sgn} \circ \varphi)$. Then $K \triangleleft G$, so $K = \{e\}$ or $K = G$. Since $G/K \cong \text{im}(\text{sgn} \circ \varphi)$ has at most 2 elements, $|G|/|K| \leq 2$. For $|G| > 2$, we cannot have $K = \{e\}$, so $K = G$. Thus $\text{sgn}(\varphi(g)) = 1$ for all g , giving $\text{im } \varphi \leq A_n$ and hence $|G| \leq n!/2$. $\square$

Remark. We have seen on Example Sheet 1 that if $|G|$ is even, then G has an element of order 2. Cauchy's theorem generalises this to arbitrary primes.

Theorem (Cauchy's Theorem). Let G be a finite group and let p be a prime dividing $|G|$. Then G has an element of order p (in fact, at least $p - 1$ such

elements).

Remark. The hypothesis that p is prime is essential: A_4 has no element of order 6 even though $6 \mid 12 = |A_4|$. The converse, however, holds for any divisor by Lagrange's theorem.

Proof. Consider $G^p = G \times G \times \cdots \times G$ (the set of p -tuples) and let

$$X = \{(a_1, a_2, \dots, a_p) \in G^p : a_1 a_2 \cdots a_p = e\}.$$

Note that if $(b, b, \dots, b) \in X$ with $b \neq e$, then $b^p = e$, so b has order p (since p is prime).

Let $H = \langle h : h^p = e \rangle \cong C_p$ (a cyclic group of order p , unrelated to G). Define an action of H on X by cyclic rotation:

$$h(a_1, a_2, \dots, a_p) = (a_2, a_3, \dots, a_p, a_1).$$

We verify this is an action:

0. *Closure:* If $a_1 \cdots a_p = e$, then $a_1^{-1} = a_2 \cdots a_p$, so $a_2 \cdots a_p a_1 = a_1^{-1} a_1 = e$.
1. The identity acts trivially by construction.
2. The composition condition holds by construction.

Now $|X| = |G|^{p-1}$, since the first $p-1$ entries can be chosen freely and the last is determined. Since $p \mid |G|$, we have $p \mid |X|$. By the orbit-stabilizer theorem, each orbit has size $|H|/|\text{stab}| = p$ or 1. Since orbits partition X and $p \mid |X|$, the number of size-1 orbits must be divisible by p . The tuple $(e, e, \dots, e)$ is one such orbit, so there are at least $p-1$ others. Each size-1 orbit has the form $(a, a, \dots, a)$ with $a^p = e$ and $a \neq e$, giving an element of order p . $\square$

6 Symmetric groups II

Remark. In this chapter, we study conjugacy classes of S_n and A_n . For S_n , the classification is straightforward: two elements are conjugate if and only if they have the same cycle type. For A_n , the situation is more subtle, because the permutation needed to conjugate one element to another may be odd and hence not in A_n .

6.1 Conjugacy classes in S_n

Remark. Recall that $\sigma, \tau \in S_n$ are conjugate if there exists $\rho \in S_n$ such that $\rho\sigma\rho^{-1} = \tau$. We first investigate the special case when σ is a k -cycle.

Proposition (Conjugation of cycles). If $(a_1 a_2 \cdots a_k)$ is a k -cycle and $\rho \in S_n$, then $\rho(a_1 \cdots a_k)\rho^{-1}$ is the k -cycle $(\rho(a_1) \rho(a_2) \cdots \rho(a_k))$.

Proof. Consider $\rho(a_i)$ acted on by $\rho(a_1 \cdots a_k)\rho^{-1}$. The three permutations send $\rho(a_i) \mapsto a_i \mapsto a_{i+1} \mapsto \rho(a_{i+1})$ (indices modulo k). Since ρ is bijective, every element of $\{1, \dots, n\}$ not in the image $\{\rho(a_1), \dots, \rho(a_k)\}$ is fixed. So the result is the k -cycle $(\rho(a_1) \rho(a_2) \cdots \rho(a_k))$. $\square$

Corollary (Cycle type conjugacy criterion). Two elements in S_n are conjugate if and only if they have the same cycle type.

Proof. Suppose $\sigma = \sigma_1\sigma_2 \cdots \sigma_\ell$, where the σ_i are disjoint cycles. Then $\rho\sigma\rho^{-1} = \rho\sigma_1\rho^{-1}\rho\sigma_2\rho^{-1} \cdots \rho\sigma_\ell\rho^{-1}$. Since conjugation preserves cycle length, $\rho\sigma\rho^{-1}$ has the same cycle type as σ .

Conversely, if σ and τ have the same cycle type, say

$\sigma = (a_1 a_2 \cdots a_k)(a_{k+1} \cdots a_{k+\ell}) \cdots, \quad \tau = (b_1 b_2 \cdots b_k)(b_{k+1} \cdots b_{k+\ell}) \cdots,$
then defining $\rho(a_i) = b_i$ for all i gives $\rho\sigma\rho^{-1} = \tau$. $\square$

Example (Conjugacy classes of S_4).

Cycle type	Example	ccl	$ C_{S_4} $	Sign
(1, 1, 1, 1)	e	1	24	+1
(2, 1, 1)	$(1\ 2)$	6	4	-1
(2, 2)	$(1\ 2)(3\ 4)$	3	8	+1
(3, 1)	$(1\ 2\ 3)$	8	3	+1
(4)	$(1\ 2\ 3\ 4)$	6	4	-1

Since a normal subgroup must be a union of conjugacy classes containing e and its order must divide 24, we can enumerate all normal subgroups:

- (i) Order 1: $\{e\}$.

- (ii) Order 4: $\{e, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\} \cong C_2 \times C_2 = V_4$. One checks that this is indeed a subgroup.
- (iii) Order 12: A_4 (normal since it is the kernel of sgn , or equivalently has index 2).
- (iv) Order 24: S_4 .

No other unions of conjugacy classes (containing e) have order dividing 24. The corresponding quotients are $S_4/\{e\} \cong S_4$, $S_4/V_4 \cong S_3 \cong D_6$, $S_4/A_4 \cong C_2$, and $S_4/S_4 = \{e\}$.

6.2 Conjugacy classes in A_n

Remark. We have seen that conjugacy classes in S_n are classified by cycle type. For A_n , the situation is more delicate. We have

$$\begin{aligned} \text{ccl}_{S_n}(\sigma) &= \{\tau \in S_n : (\exists \rho \in S_n) \tau = \rho \sigma \rho^{-1}\}, \\ \text{ccl}_{A_n}(\sigma) &= \{\tau \in A_n : (\exists \rho \in A_n) \tau = \rho \sigma \rho^{-1}\}. \end{aligned}$$

Clearly $\text{ccl}_{A_n}(\sigma) \subseteq \text{ccl}_{S_n}(\sigma)$, but the inclusion may be strict, since the permutation needed to conjugate σ to τ may be odd.

Example (Conjugacy in A_n vs S_n). The 3-cycles $(1\ 2\ 3)$ and $(1\ 3\ 2)$ are conjugate in S_3 by $(2\ 3)$, but $(2\ 3) \notin A_3$. This does not automatically mean they are not conjugate in A_3 : there might be another even permutation that conjugates them. Indeed, in A_5 the even permutation $(2\ 3)(4\ 5)$ works, but no such element exists in A_3 .

Remark. By the orbit-stabilizer theorem applied to the conjugation action,

$$\begin{aligned} |S_n| &= |\text{ccl}_{S_n}(\sigma)| \cdot |C_{S_n}(\sigma)|, \\ |A_n| &= |\text{ccl}_{A_n}(\sigma)| \cdot |C_{A_n}(\sigma)|. \end{aligned}$$

Since $|A_n| = \frac{1}{2}|S_n|$ and $\text{ccl}_{A_n}(\sigma) \subseteq \text{ccl}_{S_n}(\sigma)$, there are exactly two possibilities: either $\text{ccl}_{A_n}(\sigma) = \text{ccl}_{S_n}(\sigma)$ (and $|C_{A_n}(\sigma)| = \frac{1}{2}|C_{S_n}(\sigma)|$), or $|\text{ccl}_{A_n}(\sigma)| = \frac{1}{2}|\text{ccl}_{S_n}(\sigma)|$ (and $C_{A_n}(\sigma) = C_{S_n}(\sigma)$).

Definition (Splitting of conjugacy classes). When $|\text{ccl}_{A_n}(\sigma)| = \frac{1}{2}|\text{ccl}_{S_n}(\sigma)|$, we say that the conjugacy class of σ *splits* in A_n . Otherwise, the conjugacy class is *retained*.

Proposition (Conjugacy class splitting in A_n). For $\sigma \in A_n$, the conjugacy class of σ splits in A_n if and only if no odd permutation commutes with σ .

Proof. The conjugacy class splits if and only if the centralizer does not, i.e. $C_{A_n}(\sigma) = C_{S_n}(\sigma)$. Since $C_{A_n}(\sigma) = C_{S_n}(\sigma) \cap A_n$, the centralizer does not split if and only if $C_{S_n}(\sigma)$ contains no odd permutations. $\square$

Example (Conjugacy classes in A_4).

Cycle type	Example	$ \text{ccl}_{S_4} $	Odd element in C_{S_4} ?	$ \text{ccl}_{A_4} $
(1, 1, 1, 1)	e	1	Yes: (1 2)	1
(2, 2)	(1 2)(3 4)	3	Yes: (1 2)	3
(3, 1)	(1 2 3)	8	No	4, 4

In the (3, 1) case, $|C_{S_4}((1 2 3))| = 24/8 = 3$. Since the centralizer has odd order, it contains no odd permutations, so the conjugacy class splits.

Example (Conjugacy classes in A_5).

Cycle type	Example	$ \text{ccl}_{S_5} $	Odd element in C_{S_5} ?	$ \text{ccl}_{A_5} $
(1, 1, 1, 1, 1)	e	1	Yes: (1 2)	1
(2, 2, 1)	(1 2)(3 4)	15	Yes: (1 2)	15
(3, 1, 1)	(1 2 3)	20	Yes: (4 5)	20
(5)	(1 2 3 4 5)	24	No	12, 12

The centralizer of (1 2 3 4 5) has order $120/24 = 5$, which is odd, so the conjugacy class splits.

Lemma (Centralizer of a 5-cycle in S_5). For $\sigma = (1 2 3 4 5) \in S_5$, we have $C_{S_5}(\sigma) = \langle \sigma \rangle$.

Proof. We have $|\text{ccl}_{S_5}(\sigma)| = 24$ and $|S_5| = 120$, so $|C_{S_5}(\sigma)| = 5$. Since $\langle \sigma \rangle \subseteq C_{S_5}(\sigma)$ and $|\langle \sigma \rangle| = 5$, we conclude $C_{S_5}(\sigma) = \langle \sigma \rangle$. $\square$

Theorem (Simplicity of A_5). A_5 is simple.

Proof. Any normal subgroup of A_5 must be a union of conjugacy classes containing e , and its order must divide $|A_5| = 60$. The conjugacy class sizes are 1, 15, 20, 12, 12. No sub-collection containing 1 sums to a proper divisor of 60, so the only normal subgroups are $\{e\}$ and A_5 . $\square$

Remark. In fact, A_n is simple for all $n \geq 5$, but the proof requires further techniques (cf. IB Groups, Rings and Modules).

7 Quaternions

Remark. In the remainder of the course, we examine several important families of groups. We begin with the quaternion group.

Definition (Quaternion group). The *quaternion group* Q_8 is the set of matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \\ \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$$

which is a subgroup of $\text{GL}_2(\mathbb{C})$.

Notation. We can also write the quaternions as

$$Q_8 = \langle a, b : a^4 = e, b^2 = a^2, bab^{-1} = a^{-1} \rangle$$

Alternatively, we can write

$$Q_8 = \{1, -1, i, -i, j, -j, k, -k\}$$

with

- (i) $(-1)^2 = 1$
- (ii) $i^2 = j^2 = k^2 = -1$
- (iii) $(-1)i = -i$ etc.
- (iv) $ij = k, jk = i, ki = j$
- (v) $ji = -k, kj = -i, ik = -j$

We have

$$1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, i = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, j = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, k = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \\ -1 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, -i = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}, -j = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, -k = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$$

Lemma (Classification of order 8 groups). If G has order 8, then either G is abelian (i.e. $\cong C_8, C_4 \times C_2$ or $C_2 \times C_2 \times C_2$), or G is not abelian and isomorphic to D_8 or Q_8 (dihedral or quaternion).

Proof. Consider the different possible cases:

- If G contains an element of order 8, then $G \cong C_8$.

- If all non-identity elements have order 2, then G is abelian (cf. Example Sheet 1, Question 8). Let $a \neq b \in G \setminus \{e\}$. By the direct product theorem, $\langle a, b \rangle = \langle a \rangle \times \langle b \rangle$. Then take $c \notin \langle a, b \rangle$. By the direct product theorem, $\langle a, b, c \rangle = \langle a \rangle \times \langle b \rangle \times \langle c \rangle \cong C_2 \times C_2 \times C_2$. Since $|\langle a, b, c \rangle| = 8 = |G|$, we have $G \cong C_2 \times C_2 \times C_2$.
- G has no element of order 8 but has an order 4 element $a \in G$. Let $H = \langle a \rangle$. Since H has index 2, it is normal in G . So $G/H \cong C_2$ since $|G/H| = 2$. This means that for any $b \notin H$, bH generates G/H . Then $(bH)^2 = b^2H = H$. So $b^2 \in H$. Since $b^2 \in \langle a \rangle$ and $\langle a \rangle$ is a cyclic group, b^2 commutes with a .

If $b^2 = a$ or a^3 , then b has order 8. Contradiction. So $b^2 = e$ or a^2 .

We also know that H is normal, so $bab^{-1} \in H$. Let $bab^{-1} = a^\ell$. Since a and b^2 commute, we know that $a = b^2ab^{-2} = b(bab^{-1})b^{-1} = ba^\ell b^{-1} = (bab^{-1})^\ell = a^{\ell^2}$. So $\ell^2 \equiv 1 \pmod{4}$. So $\ell \equiv \pm 1 \pmod{4}$.

- When $\ell \equiv 1 \pmod{4}$, $bab^{-1} = a$, i.e. $ba = ab$. So G is abelian.
 - * If $b^2 = e$, then $G = \langle a, b \rangle \cong \langle a \rangle \times \langle b \rangle \cong C_4 \times C_2$.
 - * If $b^2 = a^2$, then $(ba^{-1})^2 = e$. So $G = \langle a, ba^{-1} \rangle \cong C_4 \times C_2$.
- If $\ell \equiv -1 \pmod{4}$, then $bab^{-1} = a^{-1}$.
 - * If $b^2 = e$, then $G = \langle a, b : a^4 = e = b^2, bab^{-1} = a^{-1} \rangle$. So $G \cong D_8$ by definition.
 - * If $b^2 = a^2$, then we have $G \cong Q_8$. □

8 Matrix groups

8.1 General and special linear groups

Remark. Consider $M_{n \times n}(F)$, the set of $n \times n$ matrices over a field F (such as $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{F}_p$). Matrix multiplication is associative but not, in general, commutative. To form a group, we take the identity matrix I as the identity element and restrict to invertible matrices.

Definition (General linear group $\mathrm{GL}_n(F)$). The *general linear group* is

$$\mathrm{GL}_n(F) = \{A \in M_{n \times n}(F) : A \text{ is invertible}\} = \{A \in M_{n \times n}(F) : \det A \neq 0\}.$$

Proposition ($\mathrm{GL}_n(F)$ is a group). $\mathrm{GL}_n(F)$ is a group under matrix multiplication.

Proof. The identity is $I \in \mathrm{GL}_n(F)$. The product of invertible matrices is invertible, so $\mathrm{GL}_n(F)$ is closed. Inverses exist by definition. Multiplication is associative. $\square$

Proposition (Determinant on $\mathrm{GL}_n(F)$). $\det: \mathrm{GL}_n(F) \rightarrow F \setminus \{0\}$ is a surjective group homomorphism.

Proof. We have $\det(AB) = \det A \cdot \det B$, and $\det A \neq 0$ for $A \in \mathrm{GL}_n(F)$, so $\det$ is a homomorphism. It is surjective since for any $x \in F \setminus \{0\}$, the diagonal matrix with $(1, 1)$ -entry x and all other diagonal entries 1 has determinant x . $\square$

Definition (Special linear group $\mathrm{SL}_n(F)$). The *special linear group* $\mathrm{SL}_n(F)$ is the kernel of the determinant:

$$\mathrm{SL}_n(F) = \{A \in \mathrm{GL}_n(F) : \det A = 1\}.$$

Remark. Since $\mathrm{SL}_n(F)$ is the kernel of a homomorphism, $\mathrm{SL}_n(F) \triangleleft \mathrm{GL}_n(F)$. Note also that $Q_8 \leq \mathrm{SL}_2(\mathbb{C})$.

8.2 Actions of $\mathrm{GL}_n(\mathbb{C})$

Proposition ($\mathrm{GL}_n(\mathbb{C})$ action on $\mathbb{C}^n$). $\mathrm{GL}_n(\mathbb{C})$ acts faithfully on $\mathbb{C}^n$ by left multiplication, with two orbits: $\{0\}$ and $\mathbb{C}^n \setminus \{0\}$.

Proof. We first verify the group action axioms:

- (i) If $A \in \mathrm{GL}_n(\mathbb{C})$ and $\mathbf{v} \in \mathbb{C}^n$, then $A\mathbf{v} \in \mathbb{C}^n$.
- (ii) $I\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{C}^n$.

$$(iii) \quad A(B\mathbf{v}) = (AB)\mathbf{v}.$$

To show faithfulness, note that a linear map is determined by its action on a basis. If A fixes each standard basis vector $\mathbf{e}_k$, then $A = I$ (since the columns of A are the images of the basis vectors).

For the orbits: $A\mathbf{0} = \mathbf{0}$ for all A , and since A is invertible, $A\mathbf{v} = \mathbf{0}$ if and only if $\mathbf{v} = \mathbf{0}$. So $\{\mathbf{0}\}$ is an orbit. For any $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$, there exists $A \in \mathrm{GL}_n(\mathbb{C})$ with $A\mathbf{v} = \mathbf{w}$ (cf. Vectors and Matrices), so $\mathbb{C}^n \setminus \{\mathbf{0}\}$ is a single orbit. $\square$

Remark. Similarly, $\mathrm{GL}_n(\mathbb{R})$ acts faithfully on $\mathbb{R}^n$ with two orbits.

Proposition ($\mathrm{GL}_n(\mathbb{C})$ conjugation action on matrices). $\mathrm{GL}_n(\mathbb{C})$ acts on $M_{n \times n}(\mathbb{C})$ by conjugation: $P \cdot A = PAP^{-1}$.

Proof. This follows directly from the definitions. $\square$

Remark. This action corresponds to change of basis: two matrices are conjugate if and only if they represent the same linear map with respect to different bases, with P being the change-of-basis matrix.

From Vectors and Matrices, the orbits of the conjugation action of $\mathrm{GL}_2(\mathbb{C})$ on $M_{2 \times 2}(\mathbb{C})$ are classified by Jordan normal form. Every $A \in M_{2 \times 2}(\mathbb{C})$ is conjugate to a matrix of one of the following forms:

- (i) $\begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$ with $\lambda \neq \mu$: two distinct eigenvalues.
- (ii) $\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$: a repeated eigenvalue with 2-dimensional eigenspace.
- (iii) $\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$: a repeated eigenvalue with 1-dimensional eigenspace.

There are three *types* of orbits, but infinitely many orbits (e.g. one for each scalar matrix λI).

8.3 Orthogonal groups

Remark. Recall that the transpose A^T is defined by $(A^T)_{ij} = A_{ji}$. Transposition has the following properties:

- (i) $(AB)^T = B^T A^T$.
- (ii) $(A^{-1})^T = (A^T)^{-1}$.
- (iii) $A^T A = I \Leftrightarrow A A^T = I \Leftrightarrow A^{-1} = A^T$. In this case, A is called *orthogonal*.
- (iv) $\det A^T = \det A$.

We work over $\mathbb{R}$ in this subsection; the complex analogue (unitary matrices) is treated later.

Remark. A matrix is orthogonal if and only if its columns (equivalently, rows) form an orthonormal basis of $\mathbb{R}^n$: we have $A A^T = I$ if and only if $\sum_k a_{ik} a_{jk} = \delta_{ij}$, i.e. $\mathbf{a}_i \cdot \mathbf{a}_j = \delta_{ij}$, where $\mathbf{a}_i$ is the i th column of A .

Lemma (Orthogonal matrices are isometries). For any orthogonal matrix A and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

- (i) $(A\mathbf{x}) \cdot (A\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$.
- (ii) $|A\mathbf{x}| = |\mathbf{x}|$.

Proof. Writing the dot product as matrix multiplication,

$$(A\mathbf{x})^T (A\mathbf{y}) = \mathbf{x}^T A^T A \mathbf{y} = \mathbf{x}^T I \mathbf{y} = \mathbf{x}^T \mathbf{y}.$$

Then $|A\mathbf{x}|^2 = (A\mathbf{x}) \cdot (A\mathbf{x}) = \mathbf{x} \cdot \mathbf{x} = |\mathbf{x}|^2$, so $|A\mathbf{x}| = |\mathbf{x}|$. $\square$

Remark. Orthogonal matrices are isometries, but not all isometries are orthogonal matrices. For example, translations are isometries but are not linear maps. However, every linear isometry of $\mathbb{R}^n$ is given by an orthogonal matrix.

Definition (Orthogonal group $O(n)$). The *orthogonal group* is

$$O(n) = O_n = O_n(\mathbb{R}) = \{A \in \text{GL}_n(\mathbb{R}) : A^T A = I\}.$$

Lemma ($O(n)$ is a group). The orthogonal group is a subgroup of $\text{GL}_n(\mathbb{R})$.

Proof. $O(n)$ is non-empty since $I \in O(n)$. If $A, B \in O(n)$, then $(AB^{-1})(AB^{-1})^T = AB^{-1}(B^{-1})^T A^T = AB^{-1} B A^{-1} = I$, so $AB^{-1} \in O(n)$. $\square$

Proposition (Determinant on $O(n)$). $\det: O(n) \rightarrow \{\pm 1\}$ is a surjective group homomorphism.

Proof. For $A \in O(n)$, we have $(\det A)^2 = \det(A^T A) = \det I = 1$, so $\det A = \pm 1$. Since $\det(AB) = \det A \cdot \det B$, the determinant is a homomorphism. It is surjective since $\det I = 1$ and

$$\det \begin{pmatrix} -1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} = -1.$$

□

Definition (Special orthogonal group $SO(n)$). The *special orthogonal group* is the kernel of $\det: O(n) \rightarrow \{\pm 1\}$:

$$SO(n) = SO_n = SO_n(\mathbb{R}) = \{A \in O(n) : \det A = 1\}.$$

Remark. By the first isomorphism theorem, $O(n)/SO(n) \cong C_2$. In particular, $SO(n) \triangleleft O(n)$ with index 2. Elements of $O(n) \setminus SO(n)$ are orientation-reversing transformations such as reflections.

Lemma ($O(n)$ coset decomposition). $O(n) = SO(n) \cup \begin{pmatrix} -1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} SO(n).$

Proof. Since $|O(n) : SO(n)| = 2$, there are exactly two cosets. The identity coset is $SO(n)$, and the other is determined by any element of determinant -1 . □

8.4 Rotations and reflections in $\mathbb{R}^2$ and $\mathbb{R}^3$

Lemma ($SO(2)$ as rotations). $SO(2)$ consists of all rotations of $\mathbb{R}^2$ about the origin.

Proof. Let $A \in SO(2)$, so $A^T A = I$ and $\det A = 1$. Write $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then $A^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. The condition $A^T = A^{-1}$ gives $c = -b$ and $d = a$, together with $ad - bc = 1$. Combining, we get $a^2 + c^2 = 1$, so we may write $a = \cos \theta$ and $c = \sin \theta$ for some θ . Then

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

This maps $(1, 0)$ to $(\cos \theta, \sin \theta)$ and $(0, 1)$ to $(-\sin \theta, \cos \theta)$, which is rotation by θ counterclockwise. □

Corollary (Classification of $O(2)$ elements). Any matrix in $O(2)$ is either a rotation about the origin or a reflection in a line through the origin.

Proof. If $A \in SO(2)$, then A is a rotation. Otherwise, by the coset decomposition,

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & -\cos \theta \end{pmatrix}$$

for some θ . This matrix has eigenvalues 1 and -1 , so it is a reflection in the eigenspace E_1 , which is a line through the origin. $\square$

Lemma ($SO(3)$ rotation axis). Every matrix in $SO(3)$ is a rotation about some axis.

Proof. Let $A \in SO(3)$. Since A is orthogonal, its eigenvalues λ satisfy $|\lambda| = 1$, and they multiply to $\det A = 1$. Since A is real, complex eigenvalues come in conjugate pairs $\lambda, \bar{\lambda}$ with $\lambda\bar{\lambda} = |\lambda|^2 = 1$, and the remaining eigenvalue must be real and equal to $+1$.

If all eigenvalues are real, then each is ± 1 and their product is 1, giving the possibilities $(1, 1, 1)$ and $(1, -1, -1)$. In either case, 1 is an eigenvalue.

So in all cases A has eigenvalue 1. Pick a unit eigenvector for eigenvalue 1 as the third basis vector. In the resulting orthonormal basis,

$$A = \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

since the third column is the image of this eigenvector, and orthogonality forces the third row to be $(0, 0, 1)$. The upper-left block

$$A' = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

satisfies $\det A' = 1$ and is orthogonal, so $A' \in SO(2)$. Therefore

$$A = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

in this basis, which is a rotation about the axis spanned by the eigenvector. $\square$

Lemma ($O(3)$ reflection decomposition). Every matrix in $O(3)$ is the product of at most three reflections in planes through the origin.

Remark. Since a rotation is a product of two reflections, this lemma says that every element of $O(3)$ is a rotation, a reflection, or the product of a rotation and a reflection.

Proof. By the coset decomposition, $O(3) = SO(3) \cup \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} SO(3)$.

If $A \in \text{SO}(3)$, then in some orthonormal basis $A = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$,

which is a product of two reflections:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ \sin \theta & -\cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

If A lies in the other coset, it is the product of a reflection and an element of $\text{SO}(3)$, hence a product of at most three reflections. $\square$

Remark. Elements of $\text{O}(3) \setminus \text{SO}(3)$ can always be written as a product of three reflections, but some may also be a single reflection. However, some genuinely require three, for example $-I = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

8.5 Unitary groups

Remark. For complex matrices, the appropriate analogue of orthogonality uses the *Hermitian conjugate* (or conjugate transpose), defined by $A^\dagger = (\overline{A})^T$, so that $(A^\dagger)_{ij} = \overline{A_{ji}}$. It has the following properties:

- (i) $(AB)^\dagger = B^\dagger A^\dagger$.
- (ii) $(A^{-1})^\dagger = (A^\dagger)^{-1}$.
- (iii) $A^\dagger A = I \Leftrightarrow AA^\dagger = I \Leftrightarrow A^\dagger = A^{-1}$. In this case, A is called *unitary*.
- (iv) $\det A^\dagger = \overline{\det A}$.

Definition (Unitary group $U(n)$). The *unitary group* is

$$U(n) = U_n = \{A \in \text{GL}_n(\mathbb{C}) : A^\dagger A = I\}.$$

Lemma (Determinant on $U(n)$). $\det: U(n) \rightarrow S^1$, where S^1 is the unit circle in $\mathbb{C}$, is a surjective group homomorphism.

Proof. For $A \in U(n)$, we have $1 = \det I = \det(A^\dagger A) = |\det A|^2$, so $|\det A| = 1$. Since $\det(AB) = \det A \cdot \det B$, the determinant is a homomorphism. For surjectivity, given $\lambda \in S^1$, the diagonal matrix with $(1, 1)$ -entry λ and all other diagonal entries 1 is unitary and has determinant λ . $\square$

Definition (Special unitary group $\text{SU}(n)$). The *special unitary group* $\text{SU}(n) = \text{SU}_n$ is the kernel of $\det: U(n) \rightarrow S^1$.

Remark. Unitary matrices preserve the Hermitian inner product: $(A\mathbf{x}) \cdot (A\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$.

9 More on regular polyhedra

Remark. In this section, we determine the symmetry groups of the cube and the tetrahedron.

9.1 Symmetries of the cube

Rotations

Remark. Recall that by the orbit-stabilizer theorem, the cube has $|G^+| = 24$ rotational symmetries.

Proposition (Cube rotation group). $G^+ \cong S_4$, where G^+ is the group of all rotations of the cube.

Proof. Consider G^+ acting on the 4 diagonals of the cube. This gives a group homomorphism $\varphi: G^+ \rightarrow S_4$. Rotation by $\frac{\pi}{2}$ about the axis through the centres of the top and bottom faces gives $(1\ 2\ 3\ 4) \in \text{im } \varphi$. Rotation by π about the axis through the midpoints of two opposite edges connecting diagonals 1 and 2 gives $(1\ 2) \in \text{im } \varphi$. Since $(1\ 2)$ and $(1\ 2\ 3\ 4)$ generate S_4 (cf. Example Sheet 2, Question 5d), φ is surjective. Since $|S_4| = 24 = |G^+|$, φ is an isomorphism. $\square$

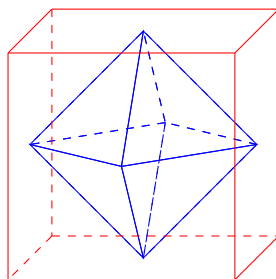
All symmetries

Remark. Consider the inversion τ through the centre of the cube, sending every point to its antipodal point. This is the map $-I$ on $\mathbb{R}^3$, so it commutes with every linear symmetry of the cube.

Proposition (Cube full symmetry group). $G \cong S_4 \times C_2$, where G is the group of all symmetries of the cube.

Proof. The inversion τ commutes with every element of G (since it suffices to check it commutes with the rotations G^+ , which generate G together with τ). We have $G^+ \cap \langle \tau \rangle = \{e\}$, so by the direct product theorem, $G^+ \times \langle \tau \rangle \hookrightarrow G$. Since $|G^+ \times \langle \tau \rangle| = 48 = |G|$, this is an isomorphism, giving $G \cong S_4 \times C_2$. $\square$

Remark. Since the octahedron is the dual of the cube (its vertices are the centres of the cube's faces), the symmetry group of the octahedron is also $S_4 \times C_2$.



9.2 Symmetries of the tetrahedron

Rotations

Proposition (Tetrahedron rotation group). $G^+ \cong A_4$, where G^+ is the group of rotational symmetries of the tetrahedron.

Proof. Label the vertices 1, 2, 3, 4. The rotation group G^+ acts on the vertices. We have $\text{orb}(1) = \{1, 2, 3, 4\}$ and $\text{stab}(1)$ consists of the rotations about the axis through vertex 1 and the centre of the opposite face, giving $|\text{stab}(1)| = 3$. By the orbit-stabilizer theorem, $|G^+| = 4 \cdot 3 = 12$.

The action on vertices gives a group homomorphism $\varphi: G^+ \rightarrow S_4$. Since a rotation fixing all four vertices must be the identity, $\ker \varphi = \{e\}$, so G^+ embeds in S_4 as a subgroup of order 12.

To identify which subgroup, note that rotation about the axis through vertex 1 gives the 3-cycles $(2\ 3\ 4)$ and $(2\ 4\ 3)$. Similarly, axes through other vertices yield all 3-cycles. Rotation by π about the axis through the midpoints of edges 1–2 and 3–4 gives the double transposition $(1\ 2)(3\ 4)$, and similarly we obtain all double transpositions. These are precisely the even permutations, so $G^+ \cong A_4$. $\square$

Remark. In particular, there is no rotation of the tetrahedron that fixes two vertices and swaps the other two, since transpositions are odd.

All symmetries

Proposition (Tetrahedron full symmetry group). $G \cong S_4$, where G is the group of all symmetries of the tetrahedron.

Proof. The plane through vertices 1, 2 and the midpoint of edge 3–4 is a plane of reflective symmetry. Reflection in this plane gives the transposition $(3\ 4)$. Now $\text{stab}(1)$ contains both $(2\ 3\ 4)$ (from a rotation) and $(3\ 4)$ (from this reflection), so $\text{stab}(1) = \langle (2\ 3\ 4), (3\ 4) \rangle \cong S_3$. By the orbit-stabilizer theorem, $|G| = 4 \cdot 6 = 24$.

The action on the four vertices gives an injective homomorphism $G \hookrightarrow S_4$, and since $|G| = 24 = |S_4|$, it is an isomorphism. This reflects the fact that every permutation of the vertices extends to a symmetry of the tetrahedron. $\square$

10 Möbius group

10.1 Möbius maps

Remark. We study maps $f: \mathbb{C} \rightarrow \mathbb{C}$ of the form $f(z) = \frac{az+b}{cz+d}$ with $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$. The condition $ad - bc \neq 0$ ensures f is non-constant, since

$$f(z) - f(w) = \frac{(ad - bc)(z - w)}{(cz + d)(cw + d)}.$$

When $c \neq 0$, the expression $f(-d/c)$ involves division by zero. To handle this, we work on the *extended complex plane* (or Riemann sphere) $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$ (cf. Vectors and Matrices), defining $f(-d/c) = \infty$.

Definition (Möbius map). A *Möbius map* is a map $f: \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$ of the form

$$f(z) = \frac{az + b}{cz + d},$$

where $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$, with $f(-d/c) = \infty$ and $f(\infty) = a/c$ when $c \neq 0$. If $c = 0$, then $f(\infty) = \infty$.

Lemma (Möbius bijectivity). Every Möbius map is a bijection $\mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$.

Proof. The inverse of $f(z) = \frac{az+b}{cz+d}$ is $g(z) = \frac{dz-b}{-cz+a}$, as one verifies by composition in both directions. $\square$

Proposition (Möbius group). The set M of all Möbius maps forms a group under composition, called the *Möbius group*.

Proof. We verify the group axioms.

(i) *Closure.* If $f_1(z) = \frac{a_1z+b_1}{c_1z+d_1}$ and $f_2(z) = \frac{a_2z+b_2}{c_2z+d_2}$, then

$$f_2 \circ f_1(z) = \frac{(a_2a_1 + b_2c_1)z + (a_2b_1 + b_2d_1)}{(c_2a_1 + d_2c_1)z + (c_2b_1 + d_2d_1)}.$$

The determinant of the composite is $(a_2a_1 + b_2c_1)(c_2b_1 + d_2d_1) - (a_2b_1 + b_2d_1)(c_2a_1 + d_2c_1) = (a_1d_1 - b_1c_1)(a_2d_2 - b_2c_2) \neq 0$.

(ii) *Identity.* The identity map $\text{id}(z) = \frac{1 \cdot z + 0}{0 \cdot z + 1}$ is a Möbius map.

(iii) *Inverses.* $f^{-1}(z) = \frac{dz-b}{-cz+a}$ is a Möbius map with determinant $da - (-b)(-c) = ad - bc \neq 0$.

(iv) *Associativity.* Composition of functions is always associative. $\square$

Remark. M is not abelian: for example, $f_1(z) = 2z$ and $f_2(z) = z + 1$ give $f_1 \circ f_2(z) = 2z + 2$ but $f_2 \circ f_1(z) = 2z + 1$.

Remark. On the Riemann sphere, ∞ is no more special than any other point. The conventions $\frac{1}{\infty} = 0$, $\frac{1}{0} = \infty$, and $\frac{a \cdot \infty}{c \cdot \infty} = \frac{a}{c}$ allow one to evaluate Möbius maps at ∞ without separate case analysis.

Remark. Since $\frac{az+b}{cz+d} = \frac{\lambda az+\lambda b}{\lambda cz+\lambda d}$ for any $\lambda \neq 0$, the coefficients a, b, c, d do not uniquely determine a Möbius map. However, a choice of a, b, c, d does uniquely determine one.

Proposition ($\text{GL}_2(\mathbb{C})$ to Möbius group homomorphism). The map $\theta: \text{GL}_2(\mathbb{C}) \rightarrow M$ sending $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \frac{az+b}{cz+d}$ is a surjective group homomorphism.

Proof. Since $ad - bc \neq 0$ for any matrix in $\text{GL}_2(\mathbb{C})$, the map θ is well-defined. It is surjective by definition of M . The closure computation above shows that $\theta(A_2) \circ \theta(A_1) = \theta(A_2 A_1)$, so θ is a homomorphism. $\square$

Lemma (Kernel of θ). $\ker \theta = Z = \{\lambda I : \lambda \in \mathbb{C} \setminus \{0\}\}$, the centre of $\text{GL}_2(\mathbb{C})$.

Proof. We need $\frac{az+b}{cz+d} = z$ for all $z \in \mathbb{C}_\infty$. Setting $z = \infty$ gives $c = 0$; $z = 0$ gives $b = 0$; $z = 1$ gives $d = a$. So $\ker \theta = \{\lambda I : \lambda \neq 0\} = Z$. $\square$

Remark. By the first isomorphism theorem, $M \cong \text{GL}_2(\mathbb{C})/Z$.

Definition (Projective general linear group $\text{PGL}_2(\mathbb{C})$). The *projective general linear group* is $\text{PGL}_2(\mathbb{C}) = \text{GL}_2(\mathbb{C})/Z$.

Remark. Since $f_A = f_B$ if and only if $B = \lambda A$ for some $\lambda \neq 0$, the restriction $\theta|_{\text{SL}_2(\mathbb{C})}: \text{SL}_2(\mathbb{C}) \rightarrow M$ is also surjective, with kernel $\{\pm I\}$. So

$$M \cong \text{SL}_2(\mathbb{C})/\{\pm I\} = \text{PSL}_2(\mathbb{C}).$$

In particular, $\text{PSL}_2(\mathbb{C}) \cong \text{PGL}_2(\mathbb{C})$.

Proposition (Möbius map generators). Every Möbius map is a composition of maps of the following three types:

- (i) Dilation/rotation: $f(z) = az$, $a \neq 0$.
- (ii) Translation: $f(z) = z + b$.

(iii) Inversion: $f(z) = 1/z$.

Proof. Let $g(z) = \frac{az+b}{cz+d} \in M$.

If $c = 0$, then $g(z) = \frac{a}{d}z + \frac{b}{d}$, which is a dilation followed by a translation:

$$z \mapsto \frac{a}{d}z \mapsto \frac{a}{d}z + \frac{b}{d}.$$

If $c \neq 0$, let $z_0 = g(\infty) = a/c$ and $h(z) = \frac{1}{z-z_0}$. Then $h \circ g$ fixes ∞ , so it is of the first type. Since $h^{-1}(w) = \frac{1}{w} + z_0$ is an inversion followed by a translation, $g = h^{-1} \circ (h \circ g)$ is a composition of maps of the three types.

Alternatively, one can verify the explicit decomposition

$$z \mapsto z + \frac{d}{c} \mapsto \frac{1}{z + \frac{d}{c}} \mapsto \frac{-(ad-bc)}{c^2(z + \frac{d}{c})} \mapsto \frac{a}{c} - \frac{ad-bc}{c^2(z + \frac{d}{c})} = \frac{az+b}{cz+d}. \quad \square$$

Remark. The decomposition of a Möbius map into elementary maps is not unique: different methods yield different compositions with the same end result.

10.2 Fixed points of Möbius maps

Definition (Fixed point). A *fixed point* of f is a point $z \in \mathbb{C}_\infty$ such that $f(z) = z$.

Remark. Any Möbius map with $c = 0$ fixes ∞ . The translation $z \mapsto z+b$ ($b \neq 0$) fixes ∞ only, whereas the dilation $z \mapsto az$ ($a \neq 0, 1$) fixes both 0 and ∞ . As we shall see, a non-identity Möbius map can have at most two fixed points.

Proposition (Three fixed points rigidity). Any Möbius map with at least 3 fixed points is the identity.

Proof. The fixed points of $f(z) = \frac{az+b}{cz+d}$ in $\mathbb{C}$ satisfy $cz^2 + (d-a)z - b = 0$. This quadratic has at most two roots, unless $c = b = 0$ and $d = a$, in which case f is the identity. $\square$

Proposition (Möbius map normal form). Every Möbius map is conjugate to $f(z) = \nu z$ for some $\nu \neq 0$, or to $f(z) = z + 1$.

Proof. The surjective homomorphism $\theta: \mathrm{GL}_2(\mathbb{C}) \rightarrow M$ sends conjugacy classes in $\mathrm{GL}_2(\mathbb{C})$ to conjugacy classes in M . By Jordan normal form, every matrix in

$\mathrm{GL}_2(\mathbb{C})$ is conjugate to one of the following types:

$$\begin{aligned} \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix} &\mapsto g(z) = \frac{\lambda}{\mu} z, \\ \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} &\mapsto g(z) = z, \\ \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} &\mapsto g(z) = z + \frac{1}{\lambda}. \end{aligned}$$

In the last case, g is also represented by $\begin{pmatrix} 1 & 1/\lambda \\ 0 & 1 \end{pmatrix}$, which is conjugate to $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ in $\mathrm{GL}_2(\mathbb{C})$. So $z + \frac{1}{\lambda}$ is conjugate to $z + 1$ in M . $\square$

Remark. For $\nu \neq 0, 1$, the map $z \mapsto \nu z$ has exactly two fixed points (0 and ∞), while $z \mapsto z + 1$ has exactly one (∞). Conjugation preserves the number of fixed points, since $f(w) = w$ if and only if $hfh^{-1}(h(w)) = h(w)$, and h is a bijection.

Proposition (Möbius map fixed point count). Every non-identity Möbius map has exactly 1 or 2 fixed points.

Proof. Let $f \in M$ with $f \neq \mathrm{id}$. By the normal form proposition, there exists $h \in M$ such that hfh^{-1} is either $z \mapsto \nu z$ (with $\nu \neq 1$) or $z \mapsto z + 1$. Now $f(w) = w$ if and only if $hfh^{-1}(h(w)) = h(w)$, so h gives a bijection between the fixed points of f and those of hfh^{-1} . The result follows since νz has 2 fixed points and $z + 1$ has 1. $\square$

10.3 Permutation properties of Möbius maps

Proposition (Three-point determination). If $f, g \in M$ agree at three distinct points of $\mathbb{C}_\infty$, then $f = g$. That is, every Möbius map is uniquely determined by its values at three points.

Proof. If $f(z_i) = g(z_i)$ for distinct z_1, z_2, z_3 , then $g^{-1} \circ f$ has three fixed points, so $g^{-1} \circ f = \mathrm{id}$ by the three fixed points rigidity proposition, giving $f = g$. $\square$

Definition (Three-transitive action). An action of G on X is called *three-transitive* if the induced action on $\{(x_1, x_2, x_3) \in X^3 : x_i \text{ pairwise distinct}\}$, given by $g(x_1, x_2, x_3) = (g(x_1), g(x_2), g(x_3))$, is transitive.

Equivalently, for any two triples (x_1, x_2, x_3) and (y_1, y_2, y_3) of distinct elements of X , there exists $g \in G$ such that $g(x_i) = y_i$ for each i .

If this g is always unique, the action is called *sharply three-transitive*.

Proposition (Sharp three-transitivity of M). The Möbius group M acts sharply three-transitively on $\mathbb{C}_\infty$.

Proof. It suffices to show that any three distinct points can be sent to $\infty, 0, 1$. Given distinct $z_1, z_2, z_3 \in \mathbb{C}_\infty$, the map

$$f(z) = \frac{(z - z_2)(z_3 - z_1)}{(z - z_1)(z_3 - z_2)}$$

sends $z_1 \mapsto \infty$, $z_2 \mapsto 0$, $z_3 \mapsto 1$. (If some $z_i = \infty$, we remove the factors involving z_i ; e.g. if $z_1 = \infty$, then $f(z) = \frac{z - z_2}{z_3 - z_2}$.)

Now given another triple w_1, w_2, w_3 of distinct points, let $g \in M$ send $w_1 \mapsto \infty$, $w_2 \mapsto 0$, $w_3 \mapsto 1$. Then $g^{-1} \circ f$ sends $z_i \mapsto w_i$. Uniqueness follows from three-point determination. $\square$

Remark. Three distinct points also uniquely determine a circle or line in $\mathbb{C}$. On the Riemann sphere, a straight line can be viewed as a circle through ∞ , but we will use the phrase “circle/line” for clarity when working in $\mathbb{C}$.

Lemma (Circle and line equation in $\mathbb{C}$). The general equation of a circle or straight line in $\mathbb{C}$ is

$$Az\bar{z} + \bar{B}z + B\bar{z} + C = 0,$$

where $A, C \in \mathbb{R}$ and $|B|^2 > AC$.

Proof. A circle $|z - w| = r$ ($r > 0$) and a line $|z - a| = |z - b|$ ($a \neq b$) can both be expanded into this form. See Vectors and Matrices for the full derivation. $\square$

Remark. Setting $A = 0$ gives a straight line. If $A \neq 0$ and $B = 0$, the equation describes a circle centred at the origin. If $C = 0$, the circle passes through 0.

Proposition (Circle preservation). Möbius maps send circles and straight lines to circles and straight lines. Equivalently, Möbius maps send circles on the Riemann sphere to circles on the Riemann sphere.

Remark. In particular, a Möbius map may send a circle to a straight line or vice versa.

Proof. Since every Möbius map is a composition of translations, dilations/rotations, and inversions, it suffices to check each type. Translations and dilations/rotations clearly preserve circles and lines. For inversion, if $w = z^{-1}$, then

$$\begin{aligned} Az\bar{z} + \bar{B}z + B\bar{z} + C &= 0 \\ \Leftrightarrow Cw\bar{w} + Bw + \bar{B}\bar{w} + A &= 0, \end{aligned}$$

which is again of the required form. $\square$

Example (Cayley transform). Consider $f(z) = \frac{z-i}{z+i}$. The real line is a circle through $0, 1, \infty$. Its image is the circle through $f(\infty) = 1$, $f(0) = -1$, and $f(1) = \frac{1-i}{1+i} = -i$, which is the unit circle. Since Möbius maps are continuous, the upper half-plane maps either to the interior or exterior of the unit circle. Evaluating at i gives $f(i) = 0$, so the upper half-plane maps to the open unit disc.

10.4 Cross-ratios

Definition (Cross-ratio). Given four distinct points $z_1, z_2, z_3, z_4 \in \mathbb{C}_\infty$, their *cross-ratio* is

$$[z_1, z_2, z_3, z_4] = g(z_4),$$

where g is the unique Möbius map sending $z_1 \mapsto \infty$, $z_2 \mapsto 0$, $z_3 \mapsto 1$. In particular, $[\infty, 0, 1, \lambda] = \lambda$ for any $\lambda \neq \infty, 0, 1$. Explicitly,

$$[z_1, z_2, z_3, z_4] = \frac{z_4 - z_2}{z_4 - z_1} \cdot \frac{z_3 - z_1}{z_3 - z_2},$$

with the obvious modifications when some $z_i = \infty$.

Remark. The cross-ratio is well-defined because M acts sharply three-transitively on $\mathbb{C}_\infty$. Different authors use different orderings of z_1, z_2, z_3, z_4 in the definition; the choice does not matter as long as one is consistent.

Lemma (Cross-ratio double transposition symmetry). For distinct $z_1, z_2, z_3, z_4 \in \mathbb{C}_\infty$,

$$[z_1, z_2, z_3, z_4] = [z_2, z_1, z_4, z_3] = [z_3, z_4, z_1, z_2] = [z_4, z_3, z_2, z_1].$$

That is, the cross-ratio is preserved by any double transposition of the entries.

Proof. Direct computation from the formula. $\square$

Proposition (Cross-ratio invariance). If $f \in M$, then $[z_1, z_2, z_3, z_4] = [f(z_1), f(z_2), f(z_3), f(z_4)]$.

Proof. Let g be the unique Möbius map with $g(z_1) = \infty$, $g(z_2) = 0$, $g(z_3) = 1$, so that $g(z_4) = [z_1, z_2, z_3, z_4] = \lambda$. Then $g \circ f^{-1}$ sends

$$f(z_1) \mapsto \infty, \quad f(z_2) \mapsto 0, \quad f(z_3) \mapsto 1, \quad f(z_4) \mapsto \lambda,$$

so $[f(z_1), f(z_2), f(z_3), f(z_4)] = g \circ f^{-1}(f(z_4)) = g(z_4) = \lambda$. $\square$

Remark. The proof shows more generally that for distinct z_1, z_2, z_3, z_4 and distinct w_1, w_2, w_3, w_4 in $\mathbb{C}_\infty$, there exists $f \in M$ with $f(z_i) = w_i$ for all i if and only if $[z_1, z_2, z_3, z_4] = [w_1, w_2, w_3, w_4]$.

Corollary (Concyclic criterion). Distinct $z_1, z_2, z_3, z_4 \in \mathbb{C}_\infty$ lie on a common circle or straight line if and only if $[z_1, z_2, z_3, z_4] \in \mathbb{R}$.

Proof. Let C be the circle/line through z_1, z_2, z_3 , and let g be the unique Möbius map with $g(z_1) = \infty$, $g(z_2) = 0$, $g(z_3) = 1$. Then $g(z_4) = [z_1, z_2, z_3, z_4]$ by definition. Since Möbius maps preserve circles and lines, $z_4 \in C$ if and only if $g(z_4)$ lies on the line through $\infty, 0, 1$, i.e. $g(z_4) \in \mathbb{R}$. $\square$

11 Projective line (non-examinable)

Definition (Projective line). For a non-zero vector $\mathbf{v} \in \mathbb{C}^2$, write $\langle \mathbf{v} \rangle = \{\lambda \mathbf{v} : \lambda \in \mathbb{C}\}$ for the 1-dimensional subspace generated by $\mathbf{v}$. The *projective line* $\mathbb{CP}^1$ is the set of all 1-dimensional subspaces of $\mathbb{C}^2$.

Proposition (Action of $\mathrm{GL}_2(\mathbb{C})$ on $\mathbb{CP}^1$). The map $A \cdot \langle \mathbf{v} \rangle = \langle A\mathbf{v} \rangle$ defines an action of $\mathrm{GL}_2(\mathbb{C})$ on $\mathbb{CP}^1$.

Proof. We check well-definedness: if $\langle \mathbf{v} \rangle = \langle \mathbf{w} \rangle$, then $\mathbf{w} = \lambda \mathbf{v}$ for some $\lambda \in \mathbb{C} \setminus \{0\}$, so $\langle A\mathbf{w} \rangle = \langle A(\lambda \mathbf{v}) \rangle = \langle \lambda(A\mathbf{v}) \rangle = \langle A\mathbf{v} \rangle$. The group action axioms are immediate. $\square$

Proposition (Kernel of the action). The kernel of this action is $Z = \{\lambda I : \lambda \in \mathbb{C} \setminus \{0\}\}$, the scalar matrices.

Proof. Suppose A fixes every line. In particular, $A\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rangle = \langle \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rangle$ forces $A\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \lambda \\ 0 \end{pmatrix}$ for some λ , and $A\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rangle = \langle \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rangle$ forces $A\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ \mu \end{pmatrix}$, so $A = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$. Now $A\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rangle = \langle \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rangle$ requires $A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \lambda \\ \mu \end{pmatrix}$ to be a scalar multiple of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$, so $\lambda = \mu$. Thus $A = \lambda I$. Conversely, every scalar matrix clearly fixes every line. $\square$

Remark. Every line in $\mathbb{C}^2$ is determined by the ratio of its coordinates: $\langle \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \rangle = \langle \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \rangle$ if and only if $v_1/v_2 = w_1/w_2$ (as elements of $\mathbb{C}_\infty$). This gives a bijection $\mathbb{CP}^1 \leftrightarrow \mathbb{C}_\infty$ via $\langle \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \rangle \leftrightarrow z_1/z_2$.

Under this identification, for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{C})$,

$$A \cdot \left\langle \begin{pmatrix} z \\ 1 \end{pmatrix} \right\rangle = \left\langle \begin{pmatrix} az + b \\ cz + d \end{pmatrix} \right\rangle \leftrightarrow \frac{az + b}{cz + d},$$

so the action of $\mathrm{GL}_2(\mathbb{C})$ on $\mathbb{CP}^1$ corresponds precisely to the Möbius group acting on $\mathbb{C}_\infty$.